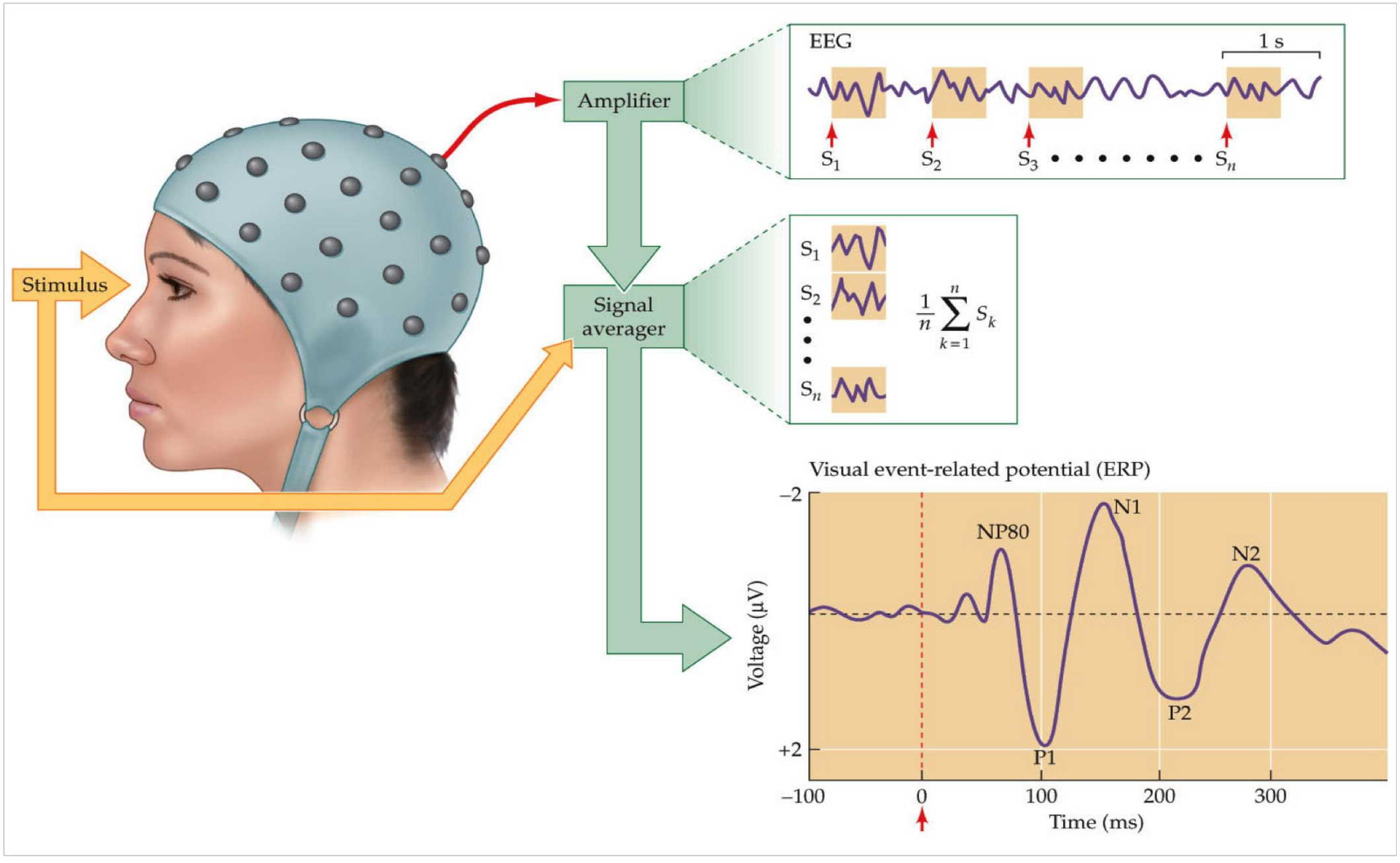
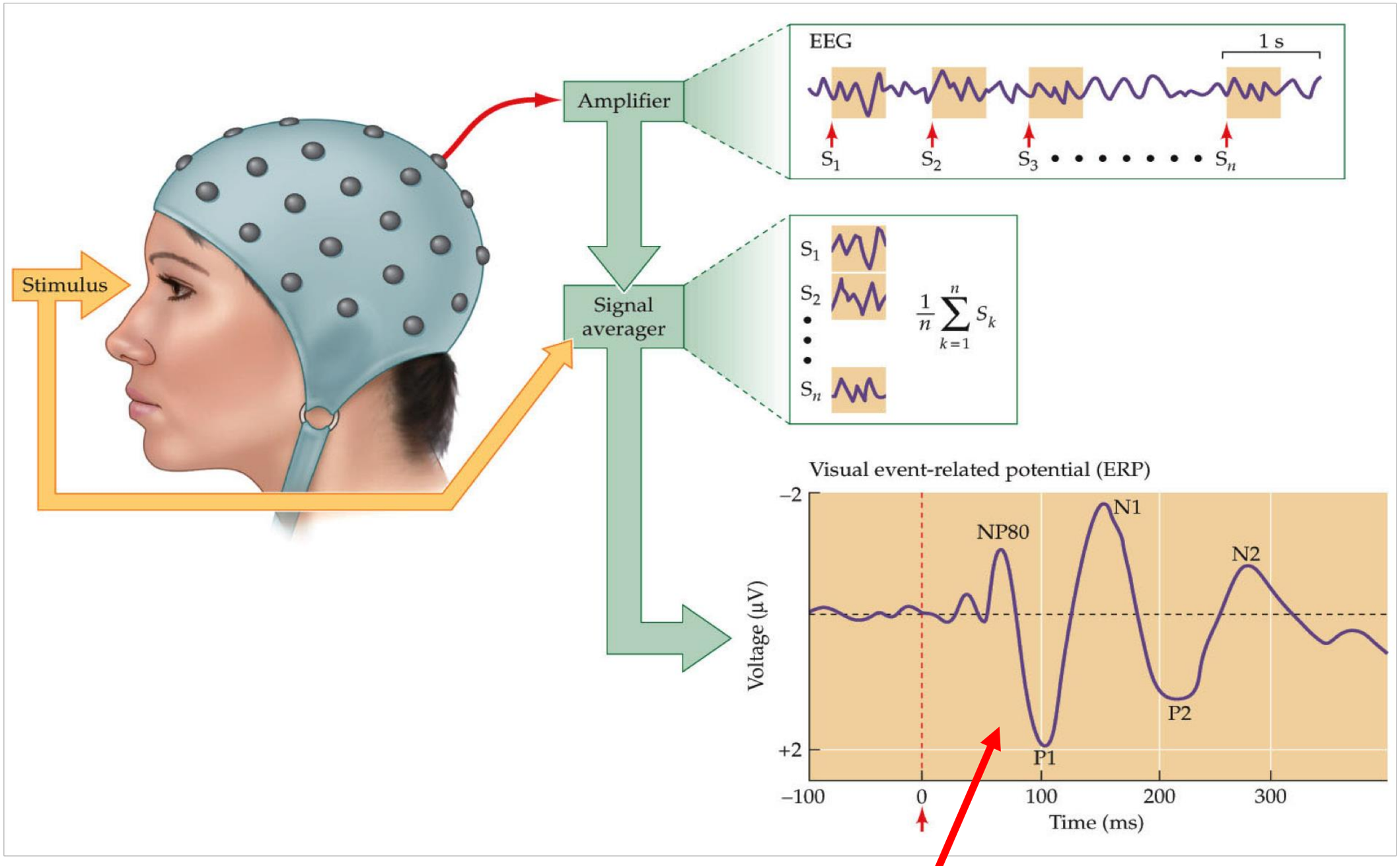
# Outline

- EEG Preprocessing
- Event-Related Potential (ERP)
- Brain Oscillation

# Event-Related Potential (ERP)



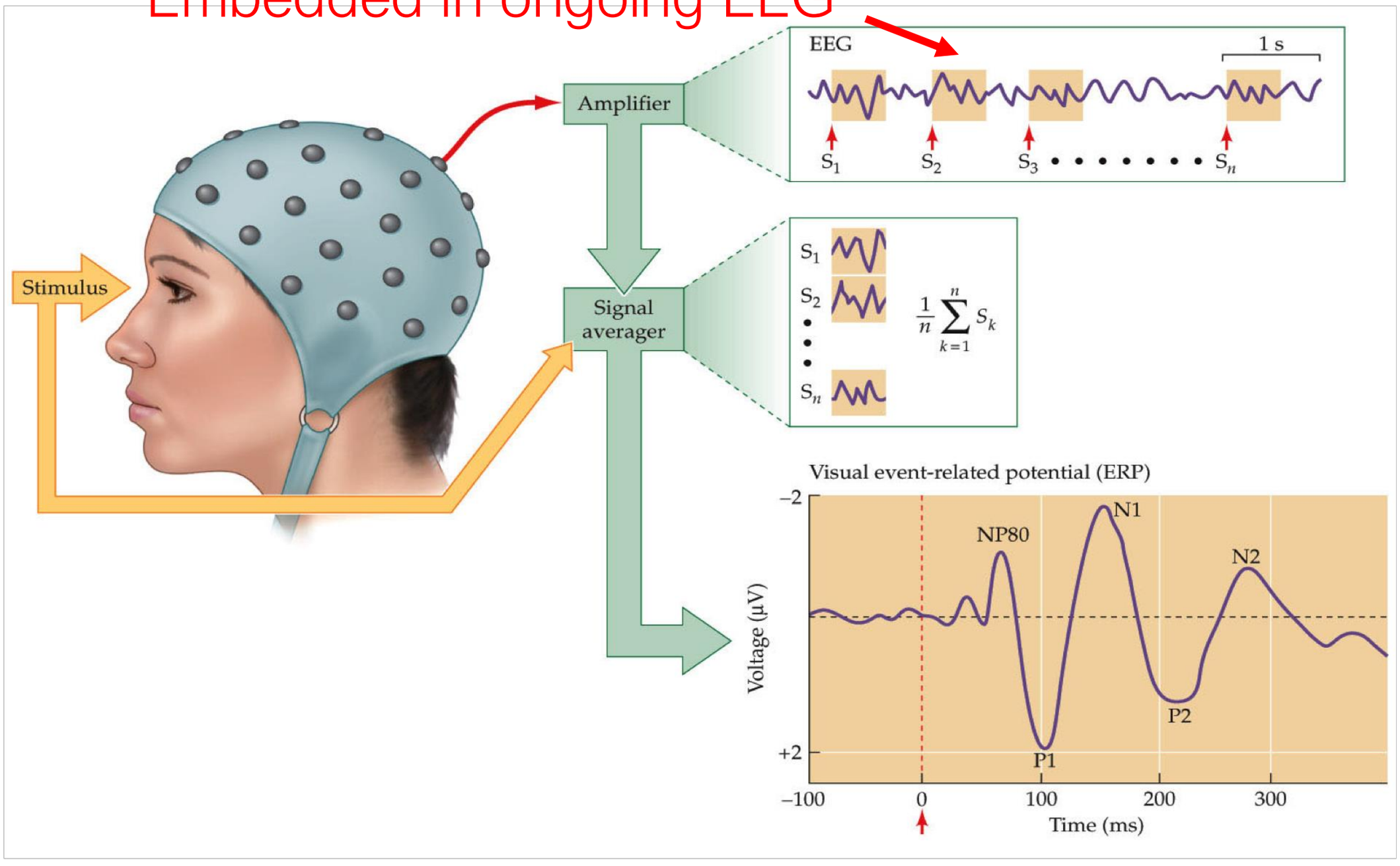
# Event-Related Potential (ERP)




Patterned voltage changes of EEG

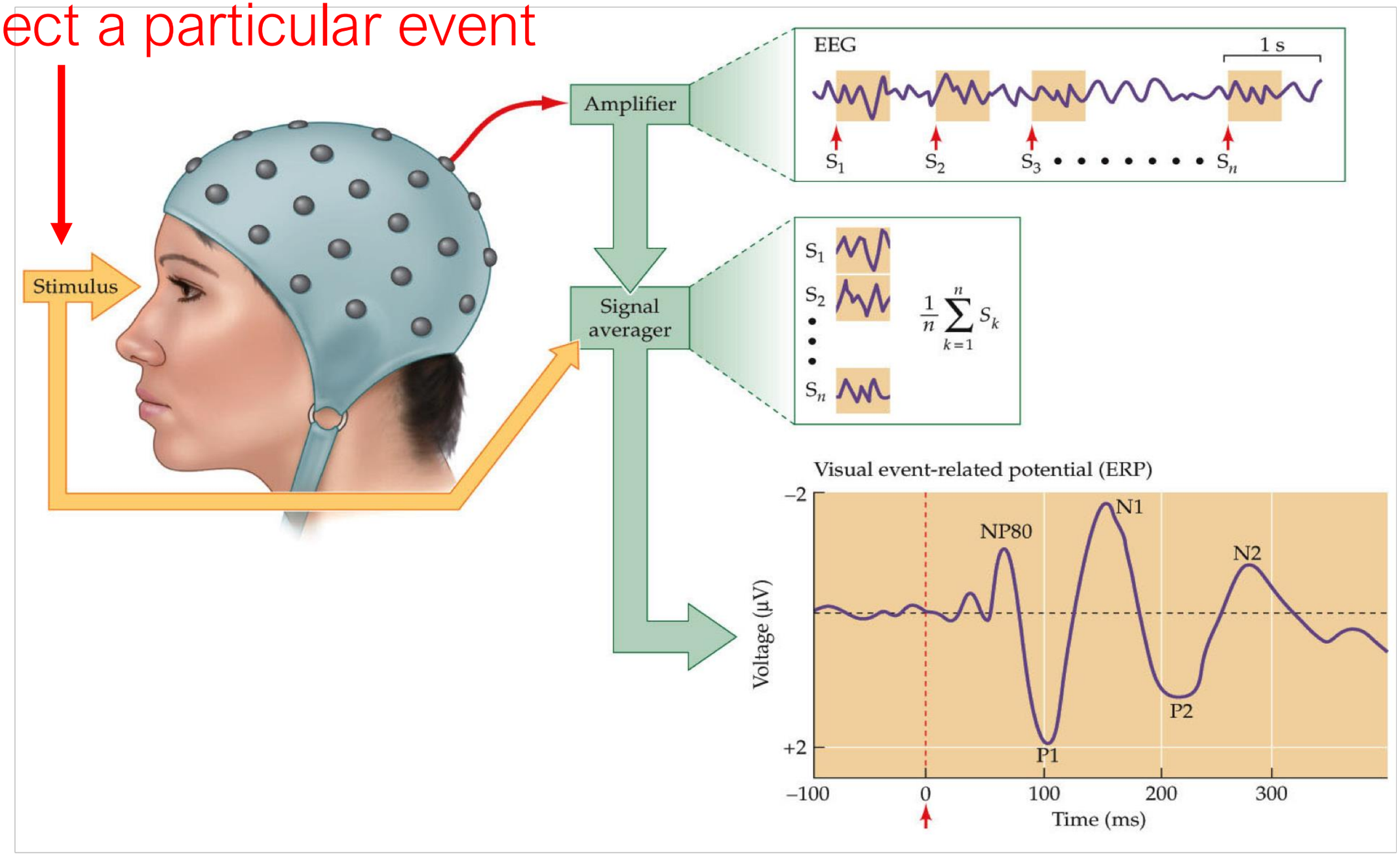
# Event-Related Potential (ERP)

Embedded in ongoing EEG



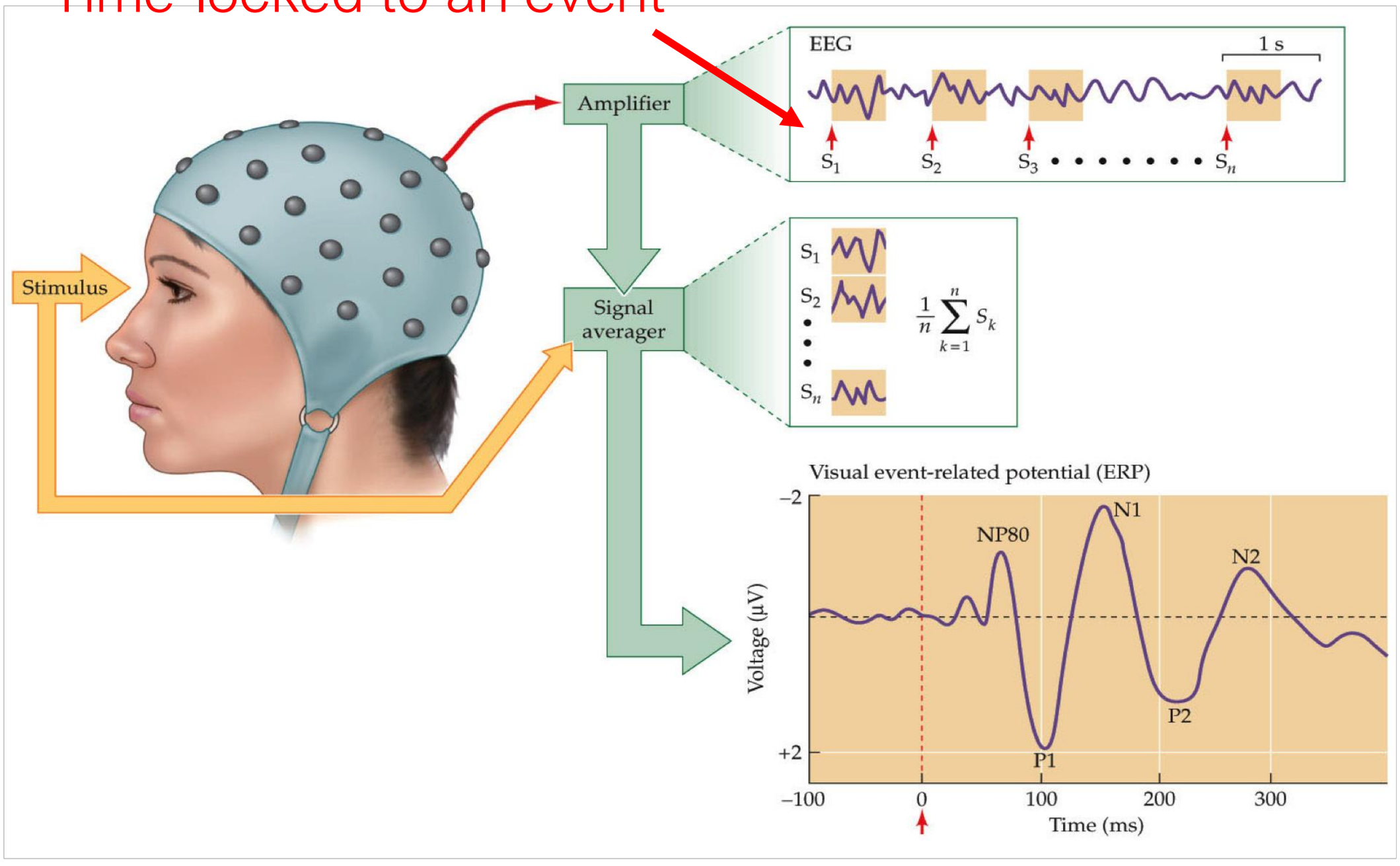
# Event-Related Potential (ERP)

Reflect a particular event



# Event-Related Potential (ERP)

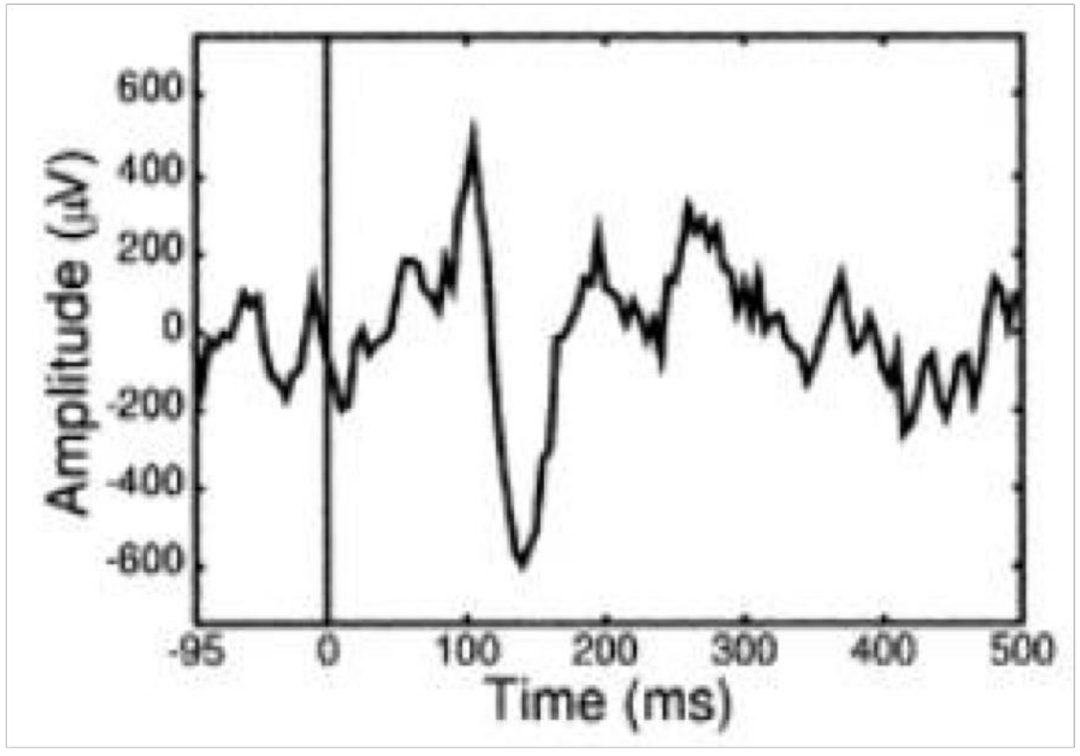
Time-locked to an event



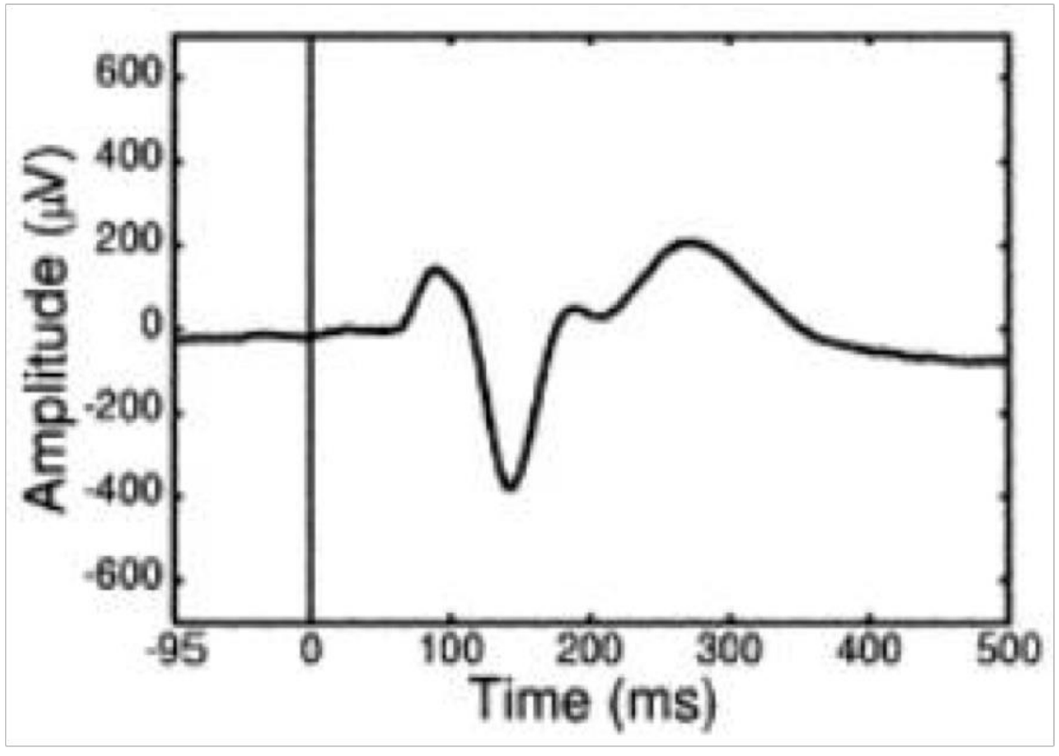
# How does averaging work for ERP?

- Stimulus-related processing is time-locked to onset
- Noise will vanish by averaging (because noise is random)

Single Trial: 100ms visual stimulus

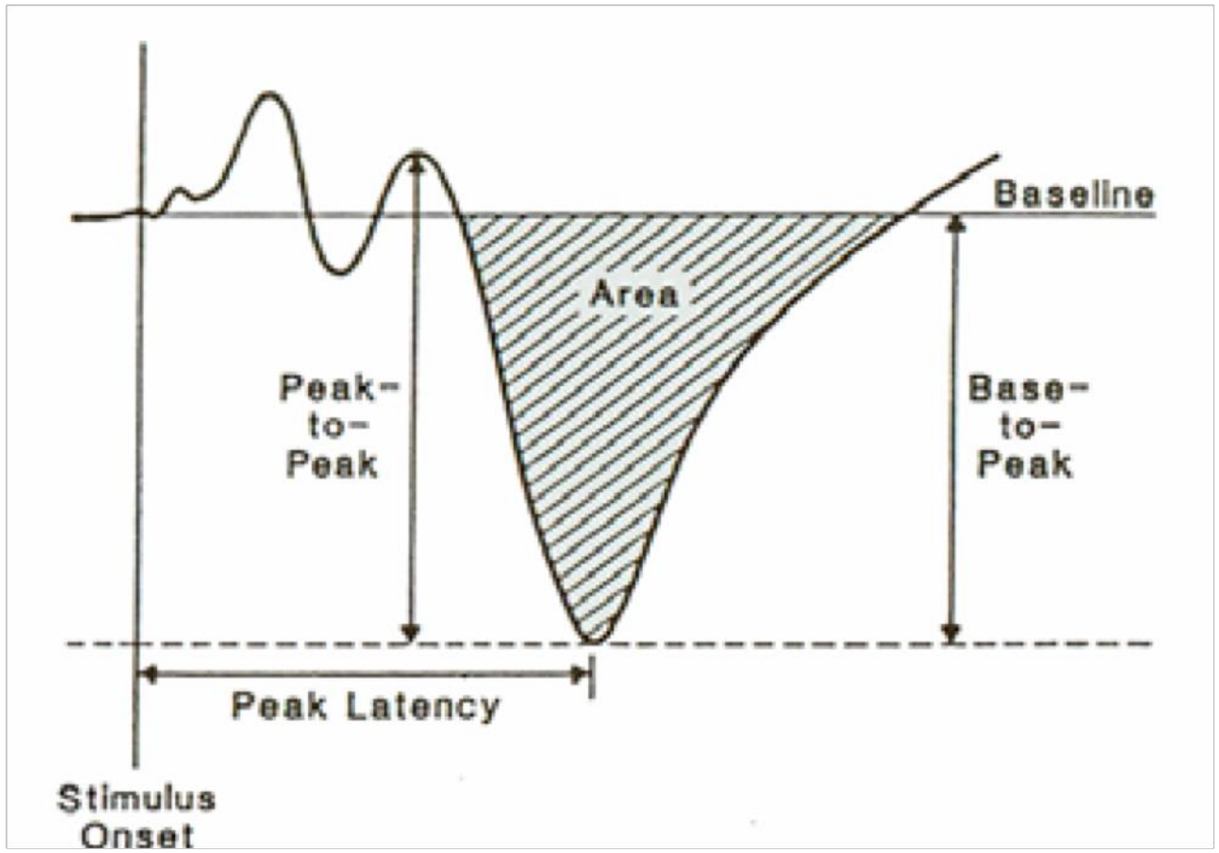



Average of 200 trials to same stimulus



# ERP Components

- Classifying ERP components
  - Polarity (positive or negative)
  - Latency
  - Area
  - Scalp locations

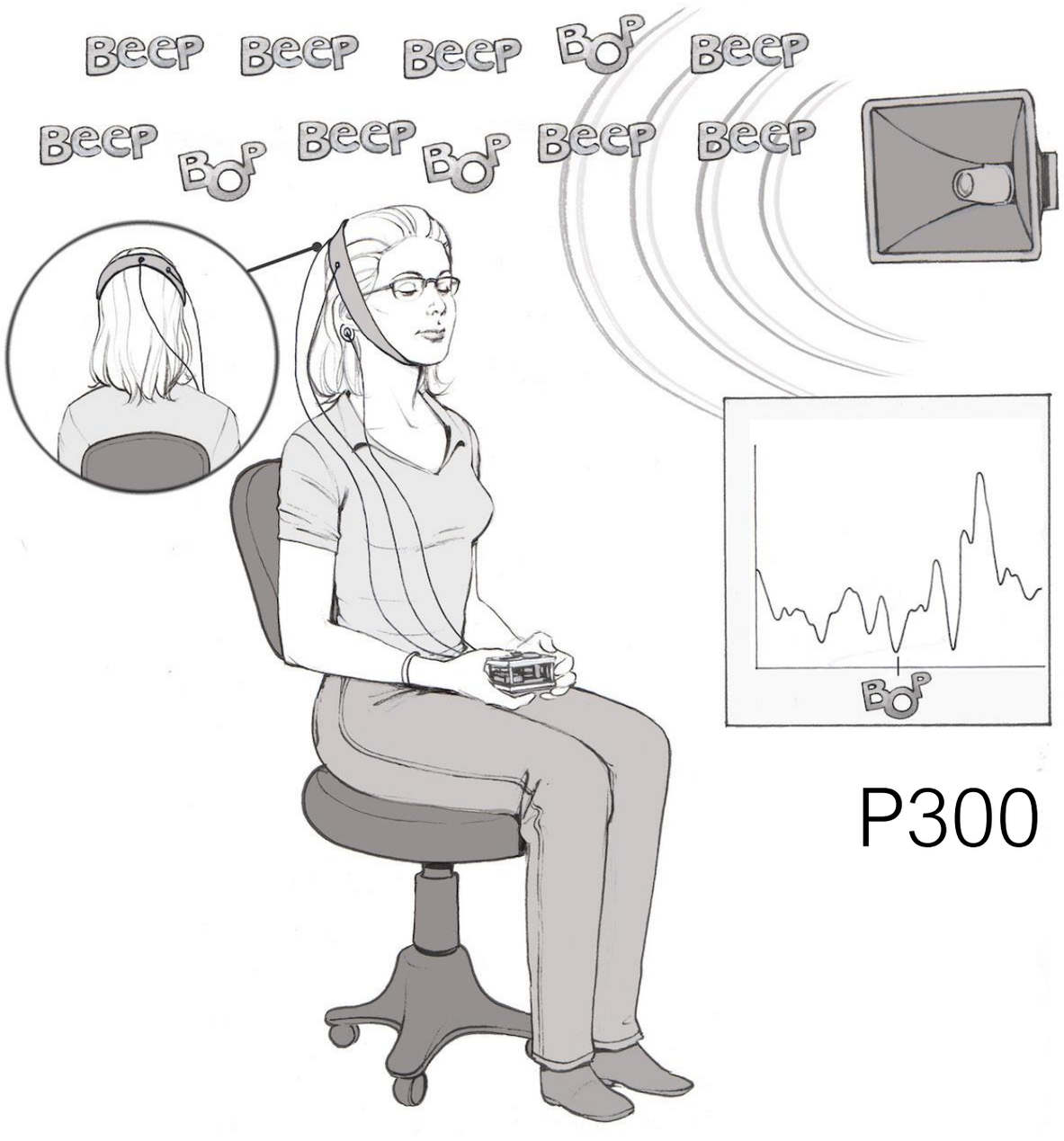


# ERP Components

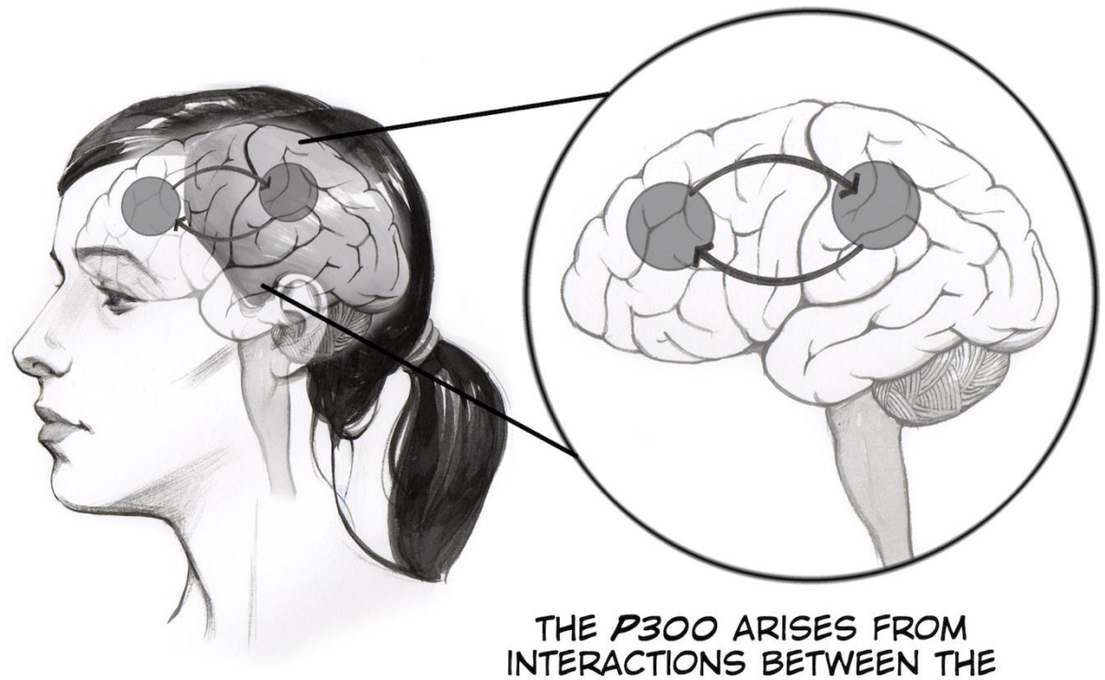
- **Early Components**
  - < 60 – 100ms
  - Occur in an obligatory fashion
  - Called **Exogenous** = determined "outside" organism
  - Even subtle deviations in appearance may be indicative of pathology

- **Late Components**
  - 100 - 500ms
  - Sensitive to the underlying brain state
  - Representing ongoing cognitive process
  - Called **Endogenous** = determined “within" organism
  - More variable in waveform and latency

# Oddball paradigm

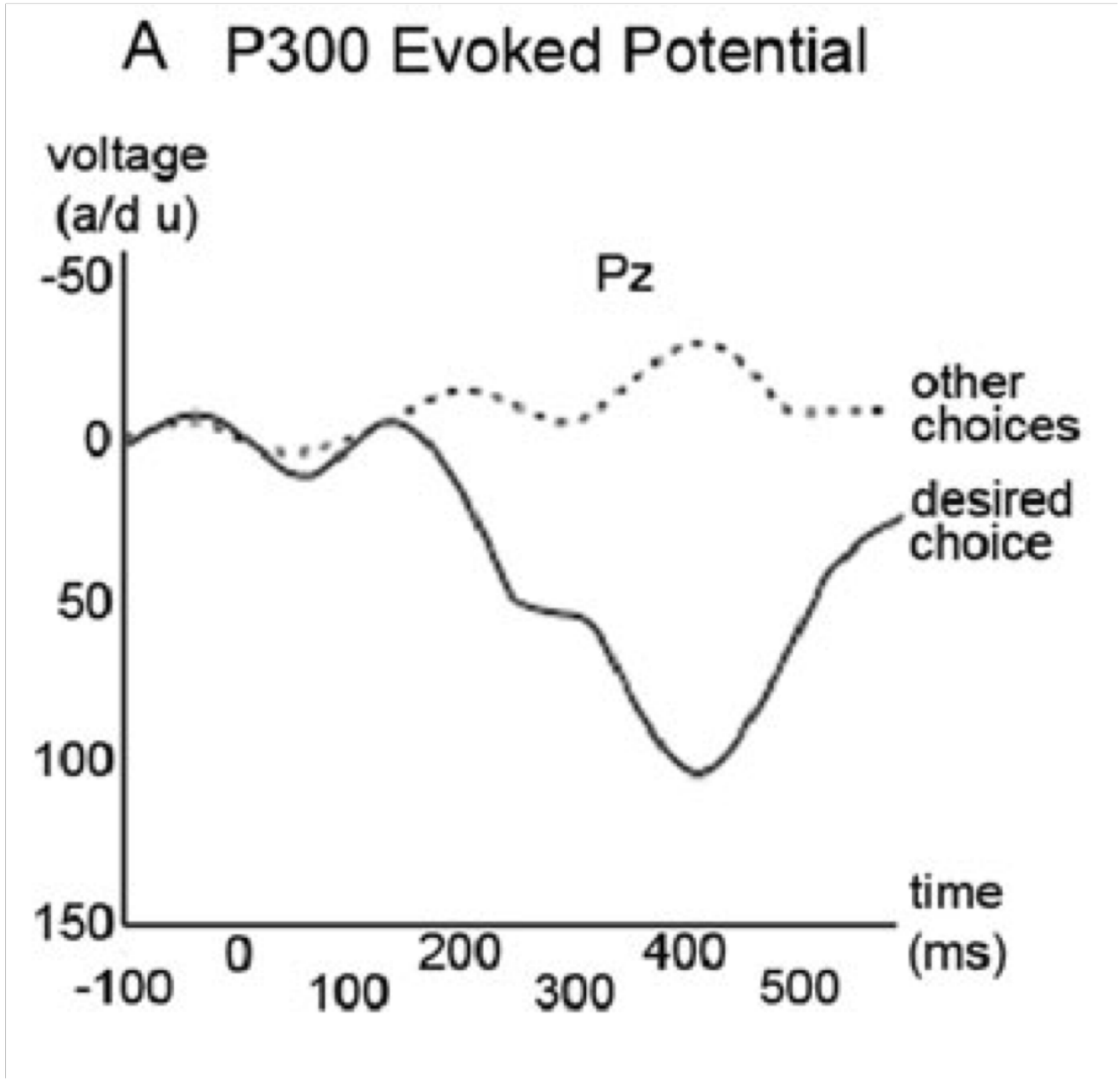



P300

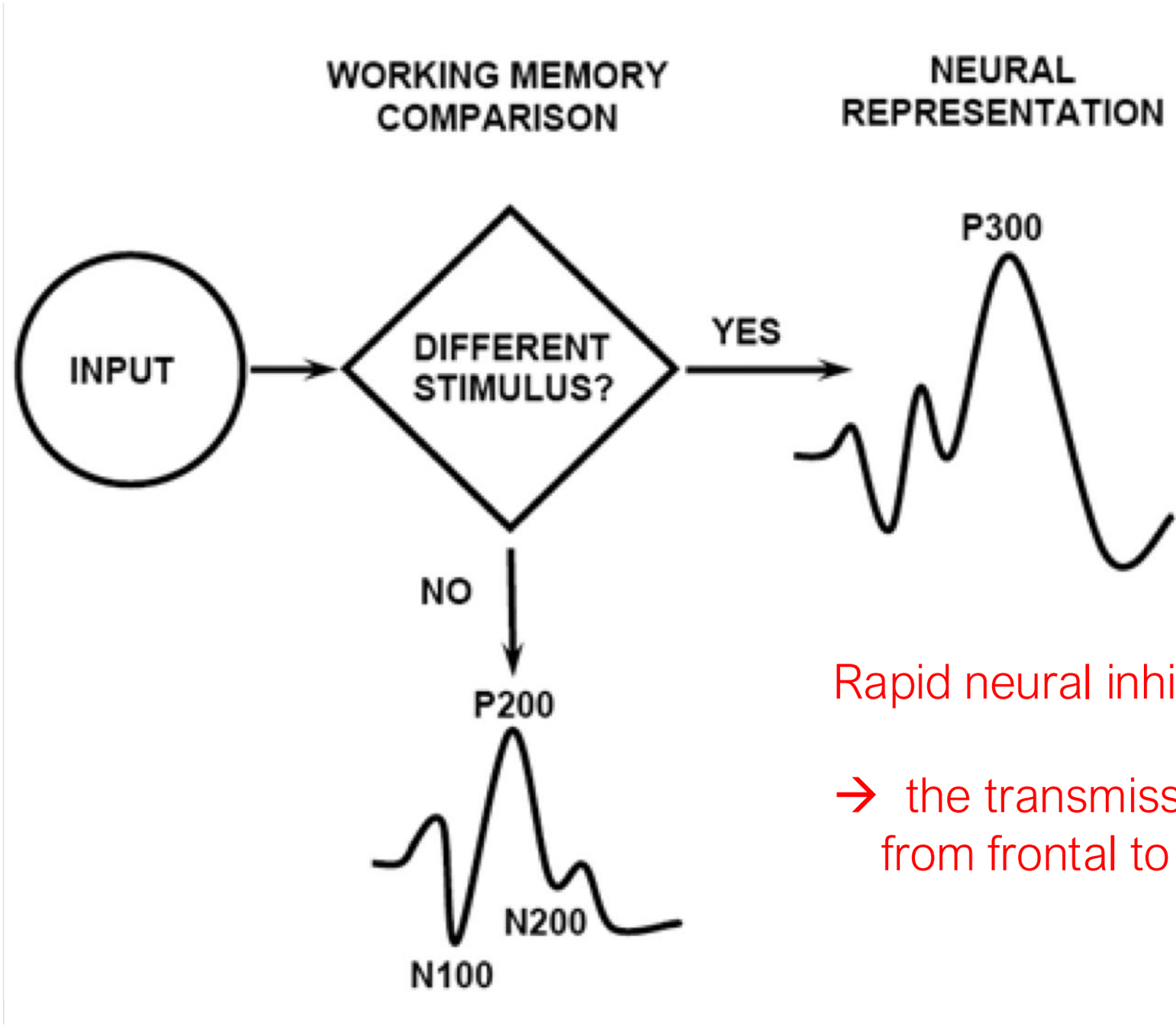


THE *P300* ARISES FROM INTERACTIONS BETWEEN THE FRONTAL AND PARIETAL LOBES

# P300 Waveform



# Origin of P300



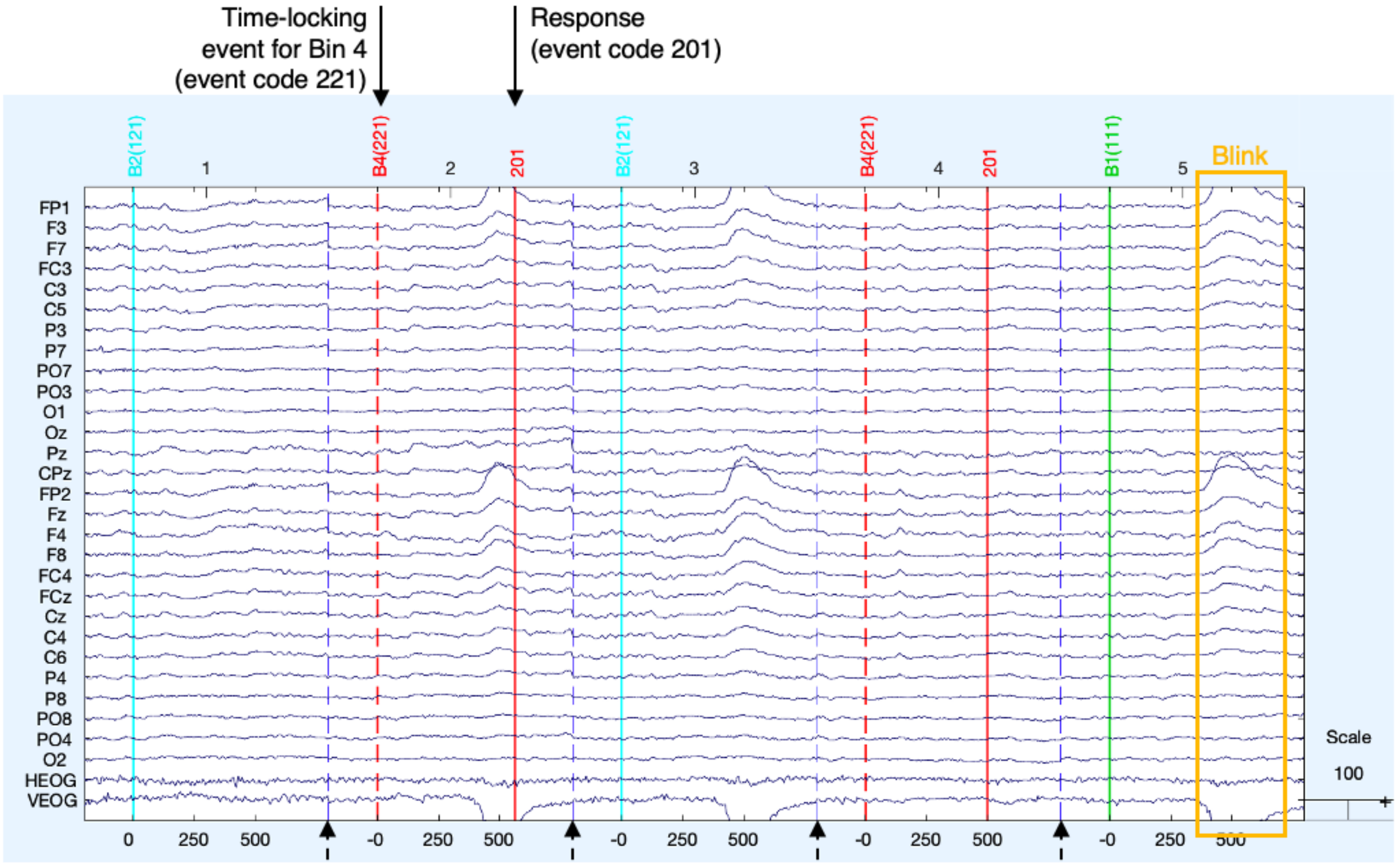

Rapid neural inhibition of ongoing activity

→ the transmission of stimulus information from frontal to temporal/parietal areas

(Linden, 2005)

# Epoch

- An epoch ~ A period in which brain responses of interest are contained
- An epoch is similar to a trial
    - Trial: related to the task design
    - Epoch: related to EEG analysis


Time-locking event for Bin 4 (event code 221)
Response (event code 201)
B2(121)
B4(221)
201
B2(121)
B4(221)
201
B1(111)
Blink
1
2
3
4
5
FP1
F3
F7
FC3
C3
C5
P3
P7
PO7
PO3
O1
Oz
Pz
CPz
FP2
Fz
F4
F8
FC4
FCz
Cz
C4
C6
P4
P8
PO8
PO4
O2
HEOG
VEOG
Scale
100
0
250
500
-0
250
500
-0
250
500
-0
250
500
-0
250
500
Epoch 1
Bin 2- Prime
(unrelated)
Epoch 2
Bin 4- Target
(unrelated)
Epoch 3
Bin 2- Prime
(unrelated)
Epoch 4
Bin 4- Target
(unrelated)
Epoch 5
Bin 1- Prime
(related)
Boundaries between epochs

(from S.J. Luck)

# Baseline correction

- Subtracting the average voltage measured during a prestimulus window (e.g., −200 to 0 ms relative to stimulus onset) from the entire waveform in an epoch

- Typical baseline duration > 100 ms

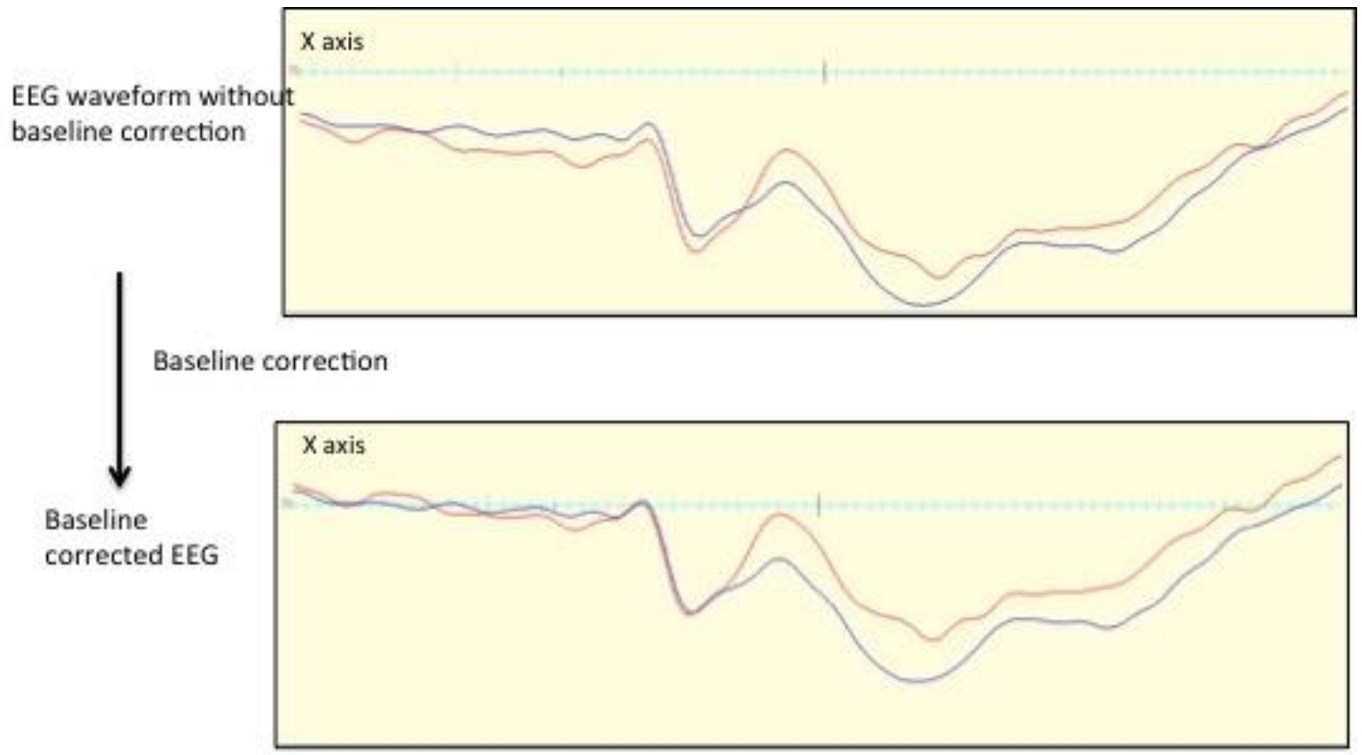



Figure I-5: baseline correction

(from S. Linnert)

# Peak and Latency

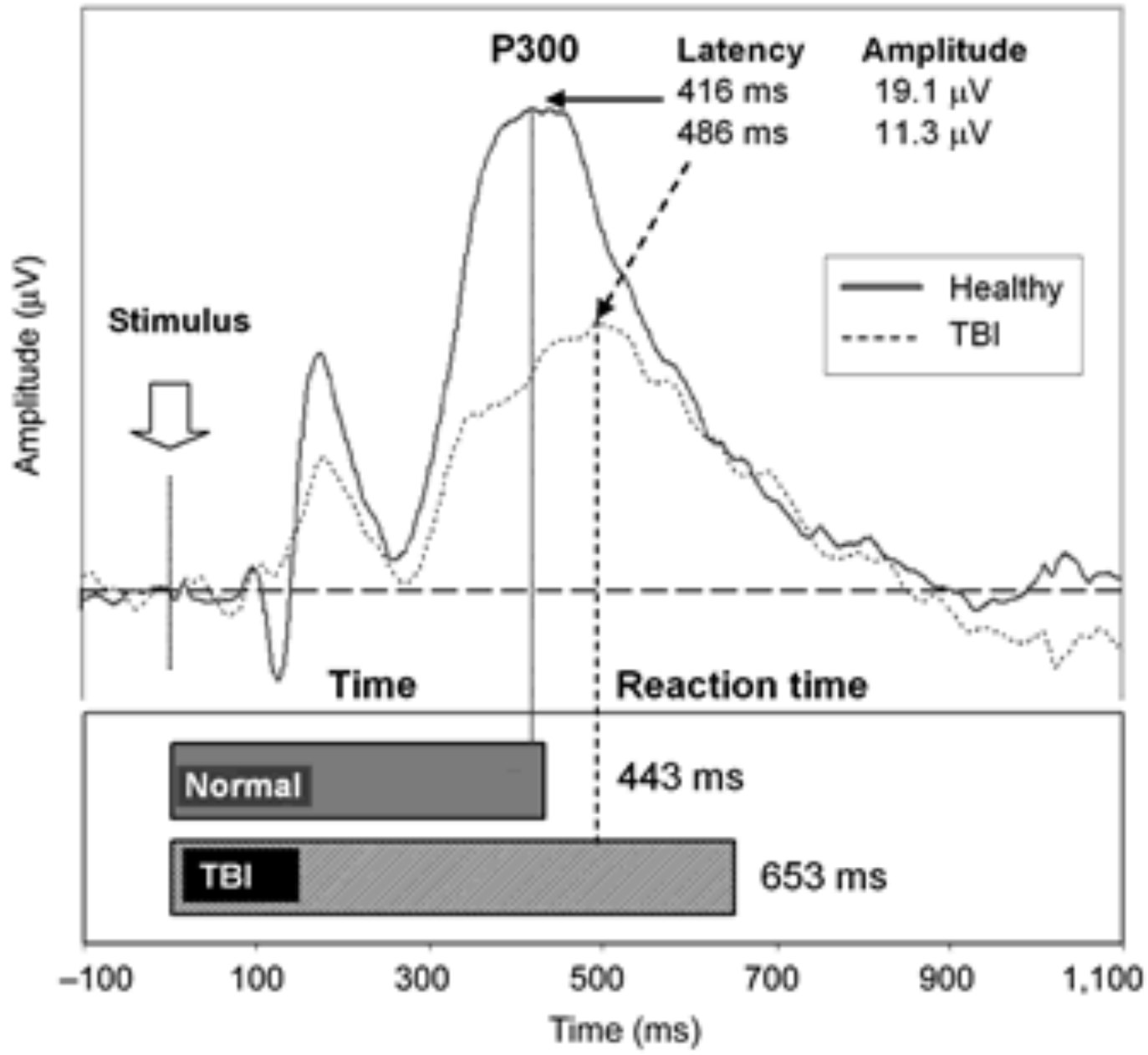
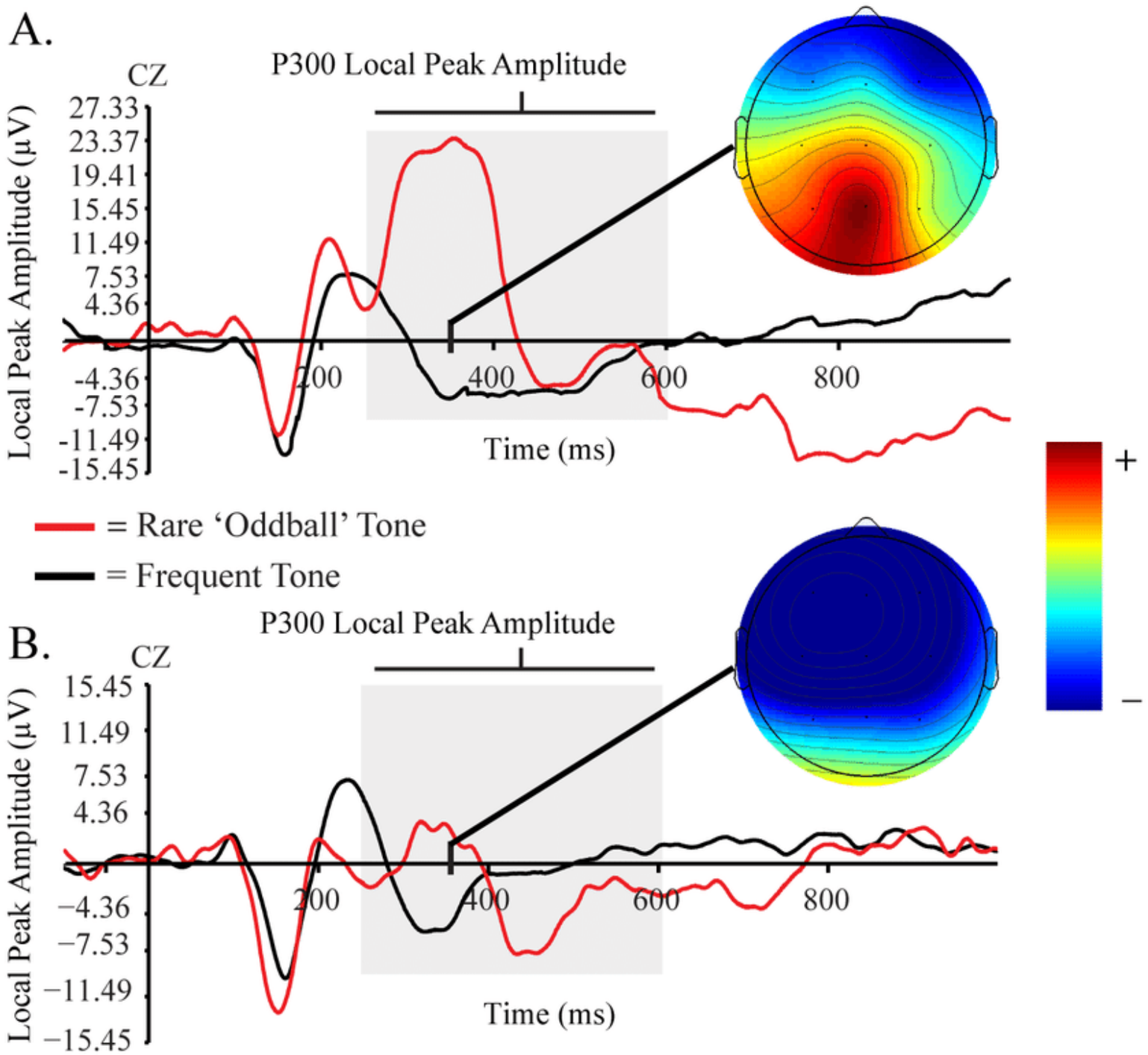


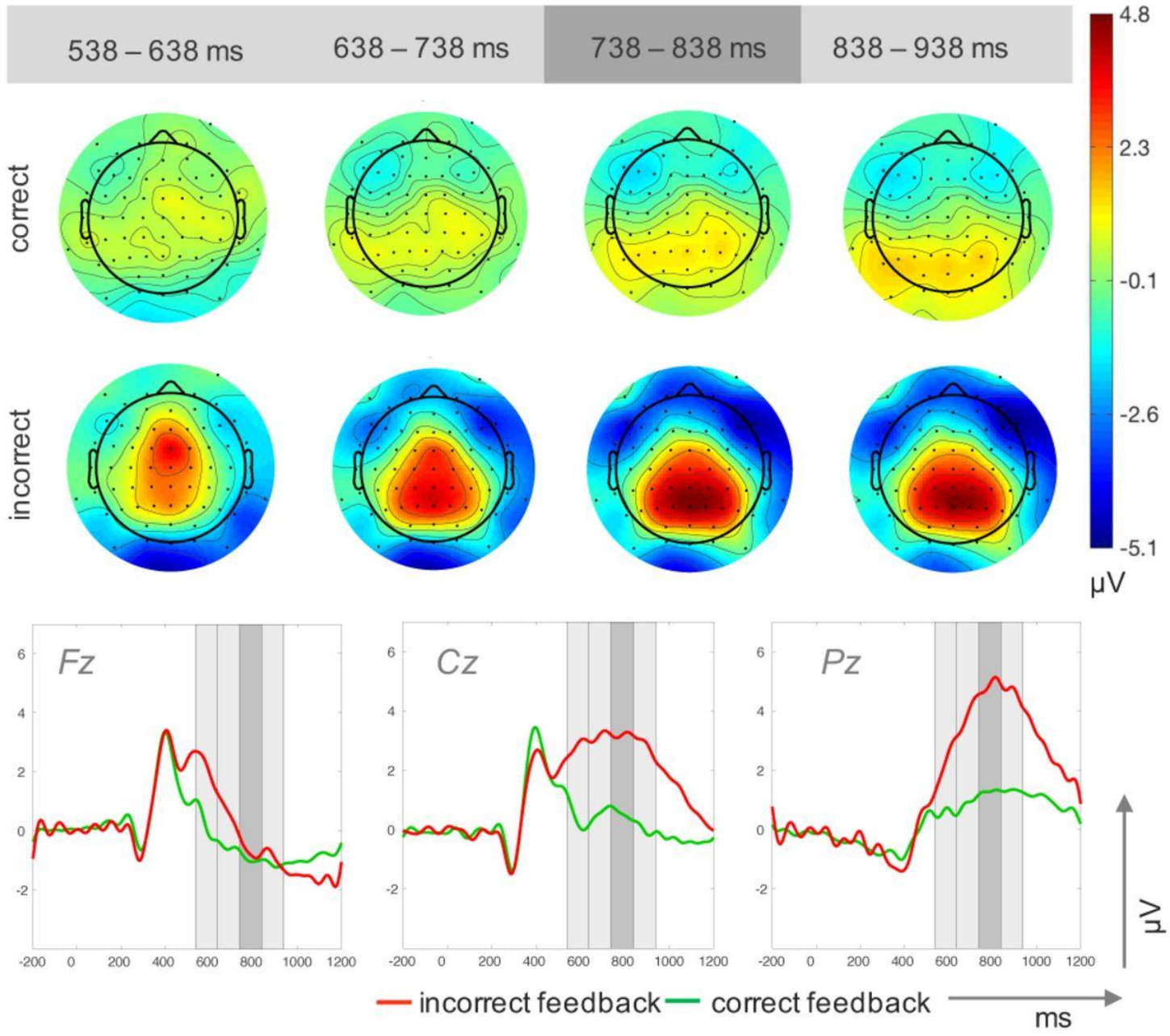

**Figure.**
Grand-average event-related potential (ERP) waveforms and reaction times. TBI = traumatic brain injury.

# ERP Topography



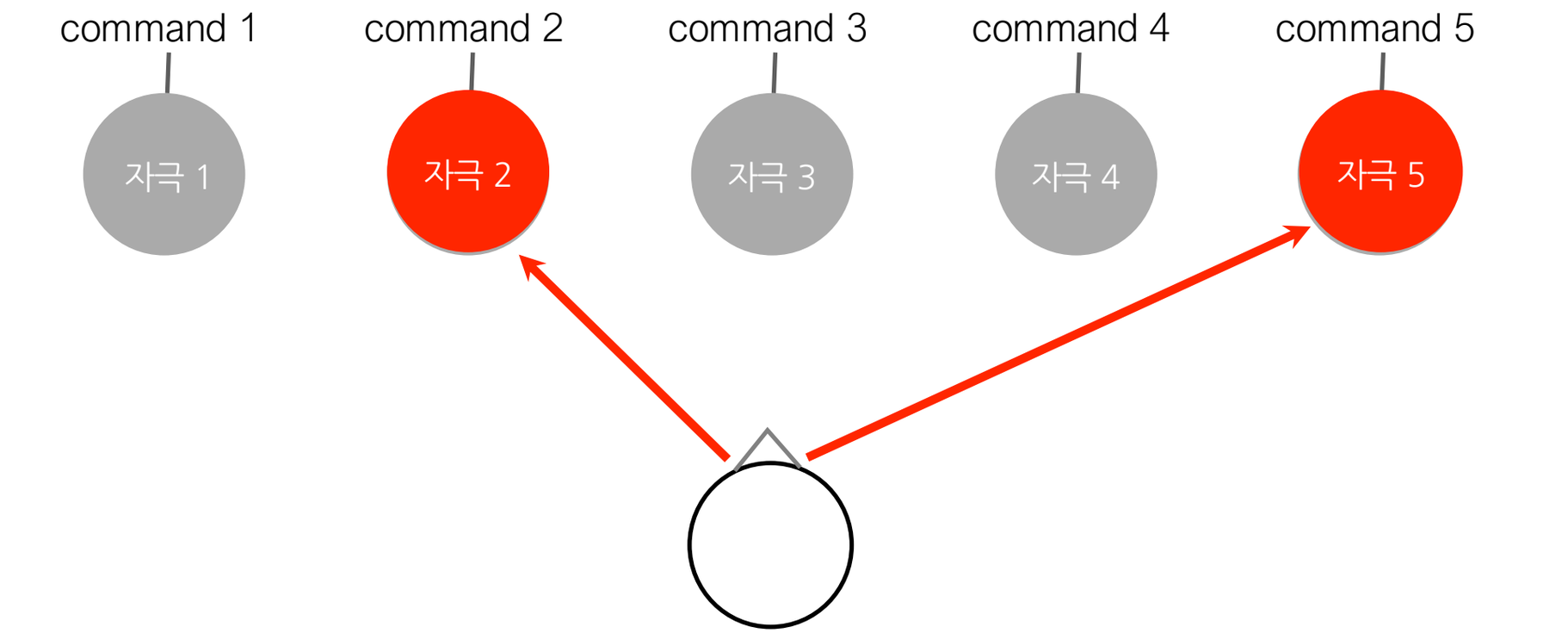

(Fercho et al., 2016)

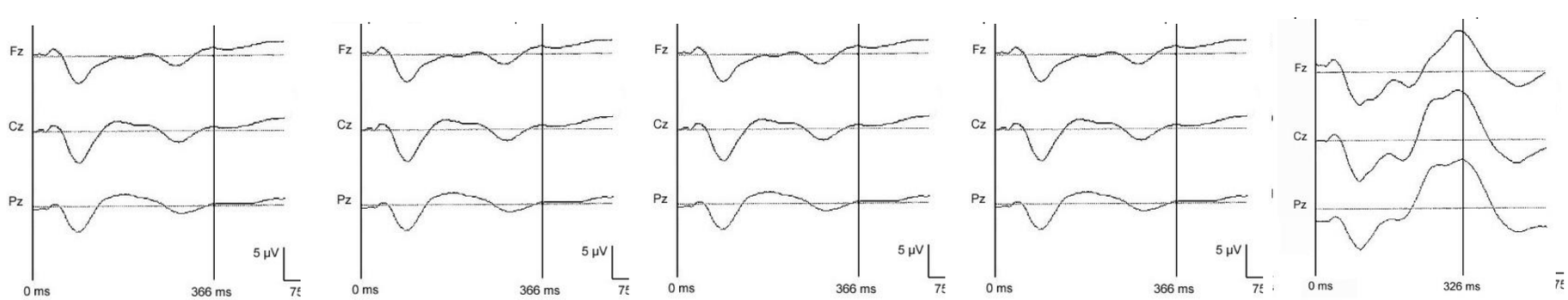
# ERP Topography



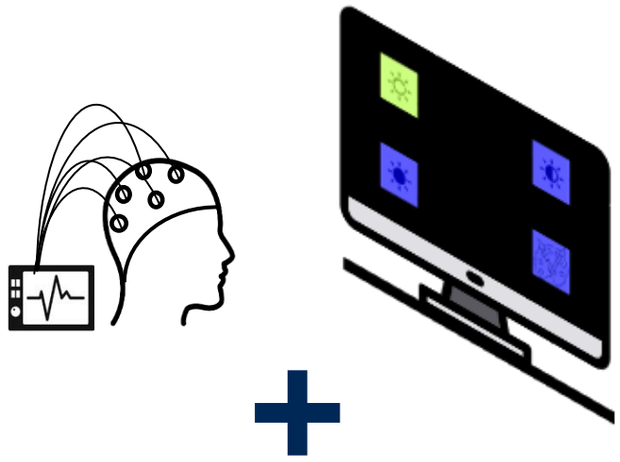

(Protzak and Granmann, 2023)

# Case Study: P300 based Brain-Computer Interface (BCI)



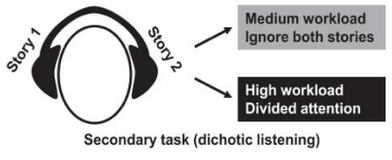



# BCI Control + Other Tasks



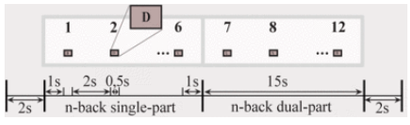
+

Dichotic listening task



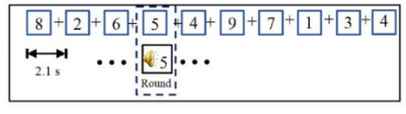

(Kathner et al., 2014)

N-back task



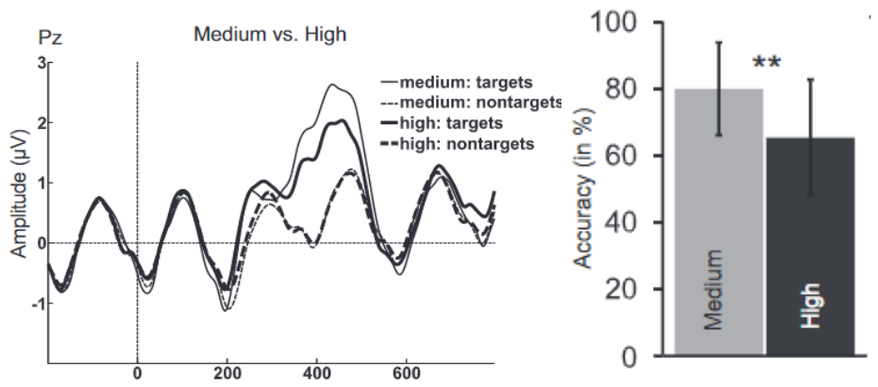

(Thurlings et al., 2013)

Mental arithmetic task

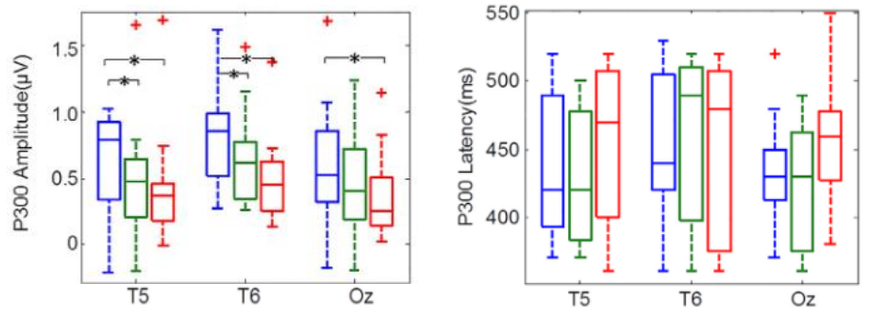



(Chen et al., 2017)

Decreased P300 amplitude and accuracy




(Kathner et al., 2014)

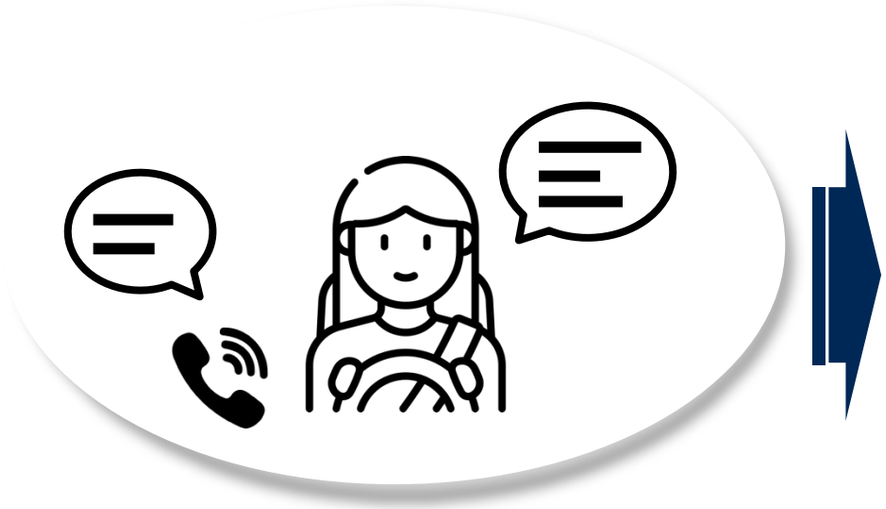



(Chen et al., 2017)

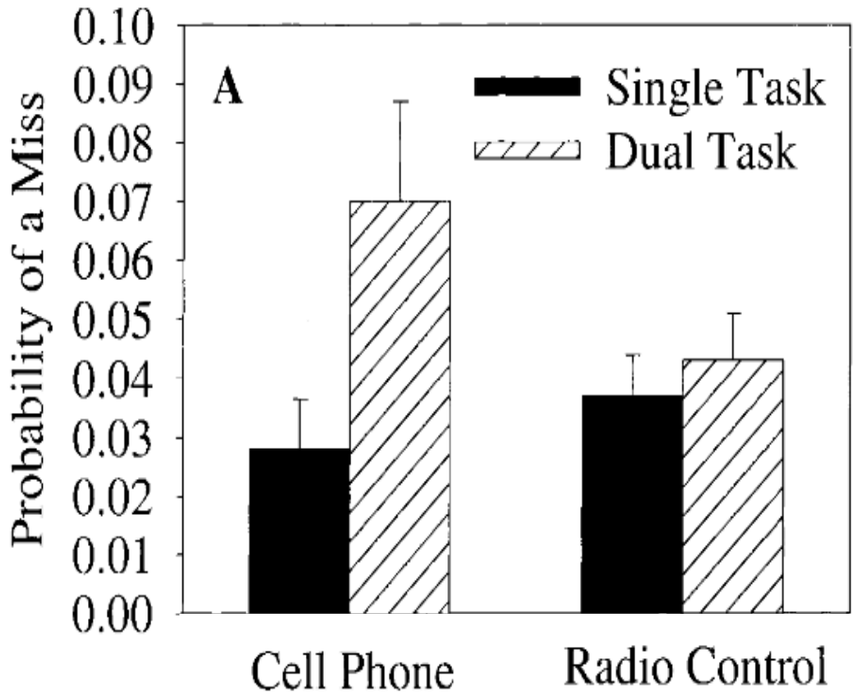
But, less likely to come across in daily life

# Real-life activities

Conversation



The driving performance decreased

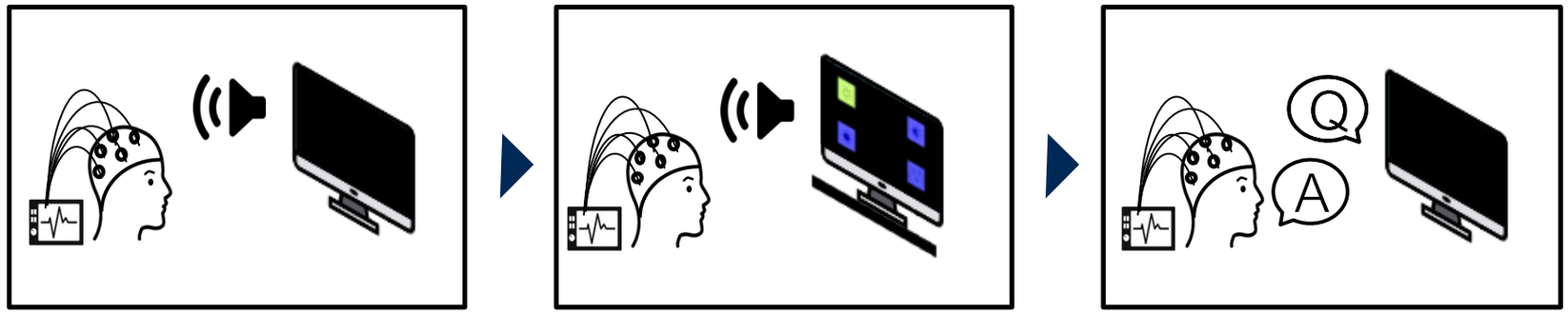



Strayer & Johnstoon, 2001

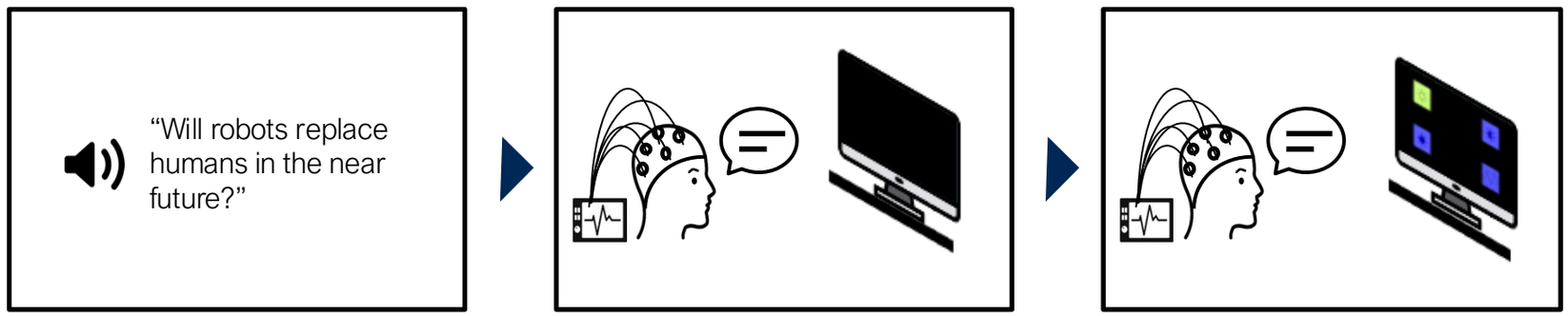
**Listening and Speaking situation might distract attention**

# Experiment

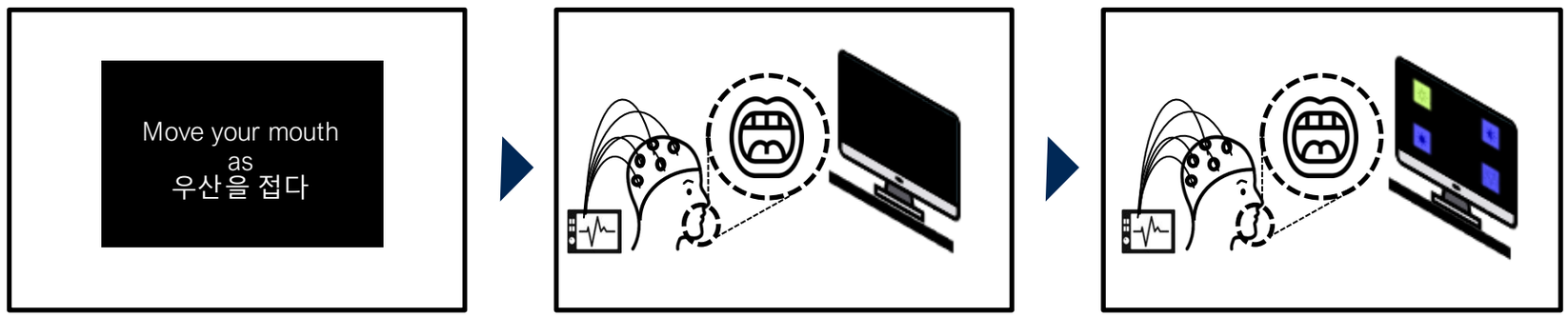
Listening




Speaking

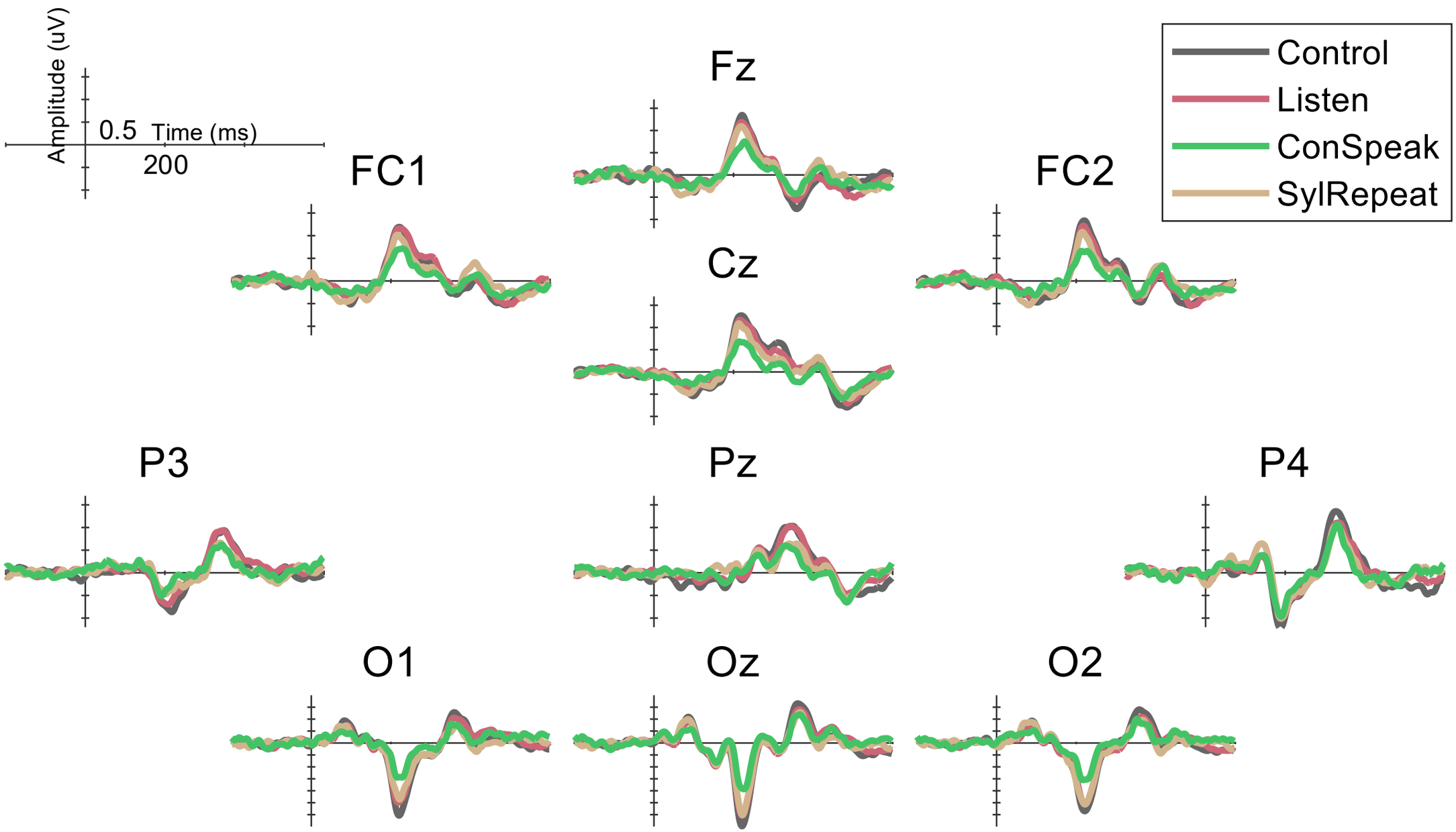



Mouth Movement



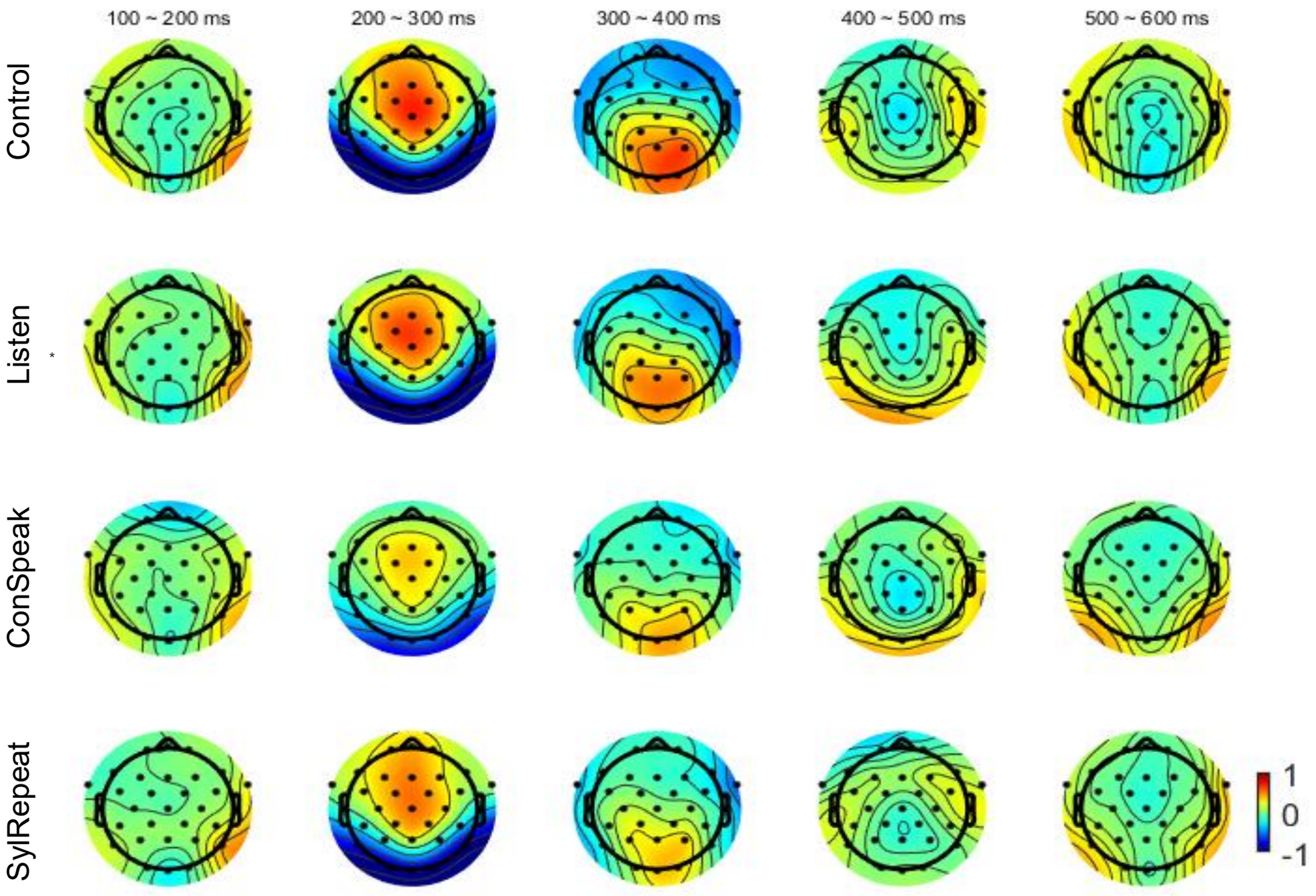

(Kim & Kim, 2024)

# ERPs



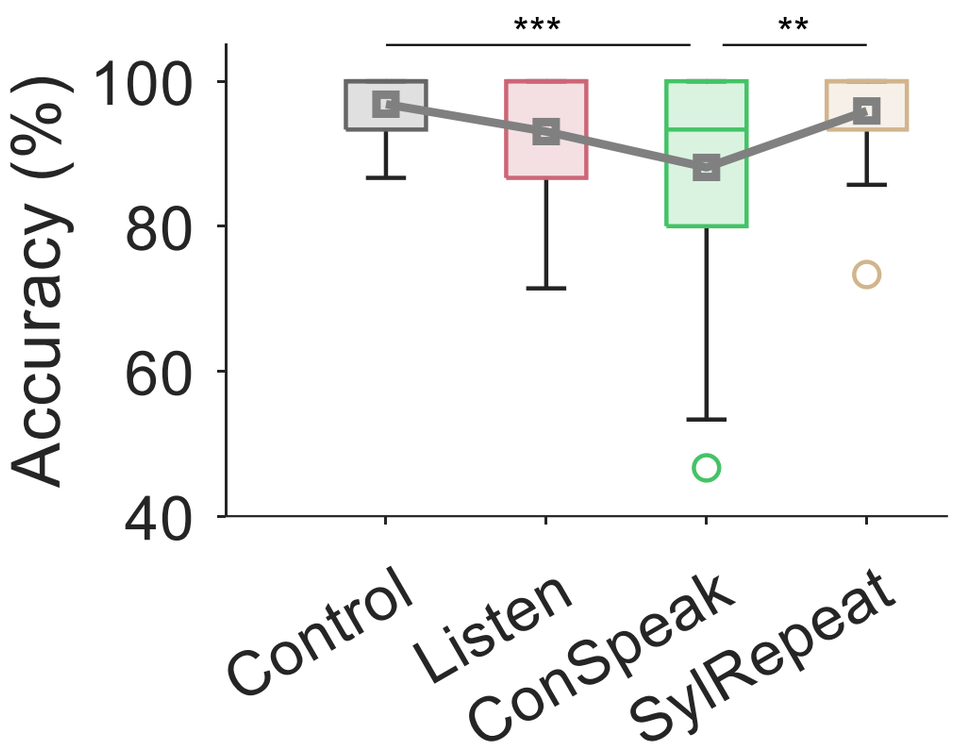

(Kim & Kim, 2024)

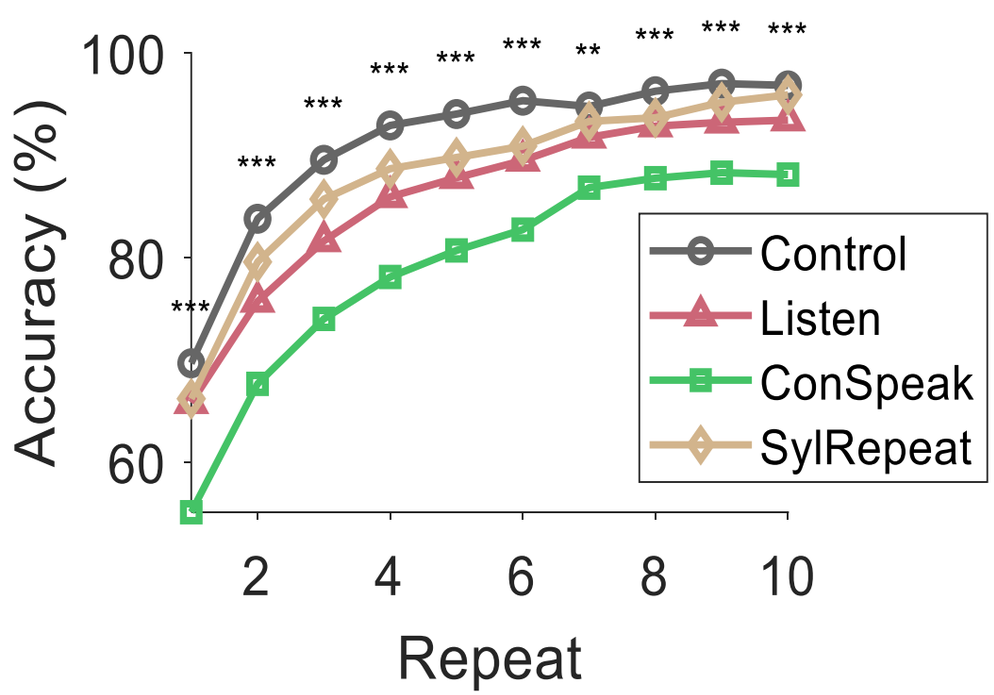
# Topography of target ERPs



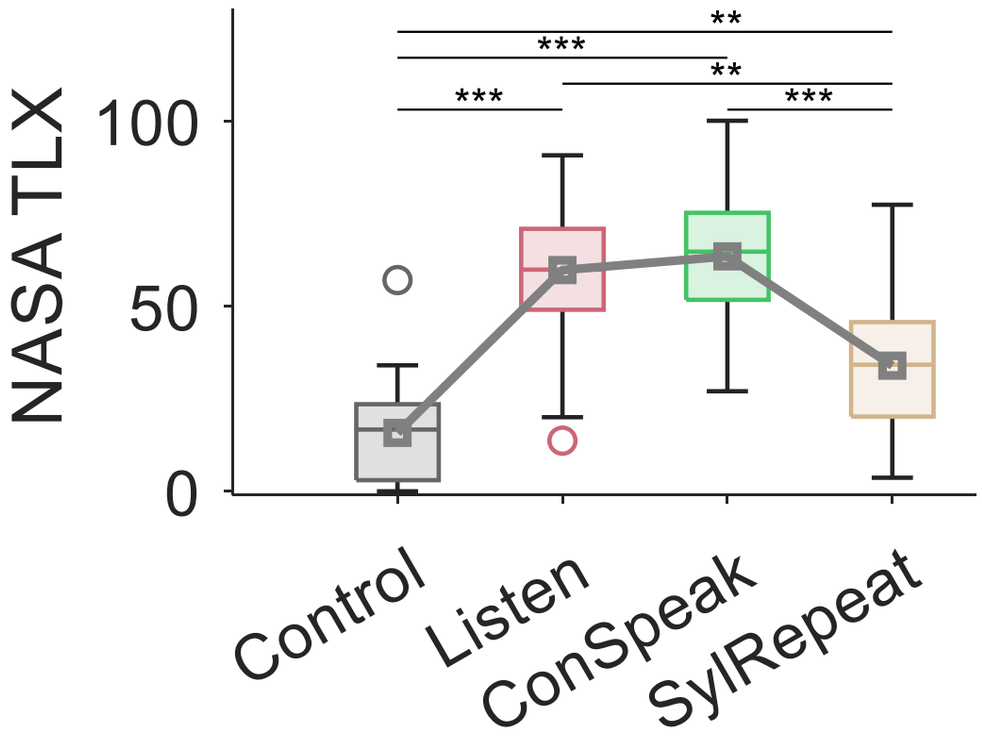

(Kim & Kim, 2024)

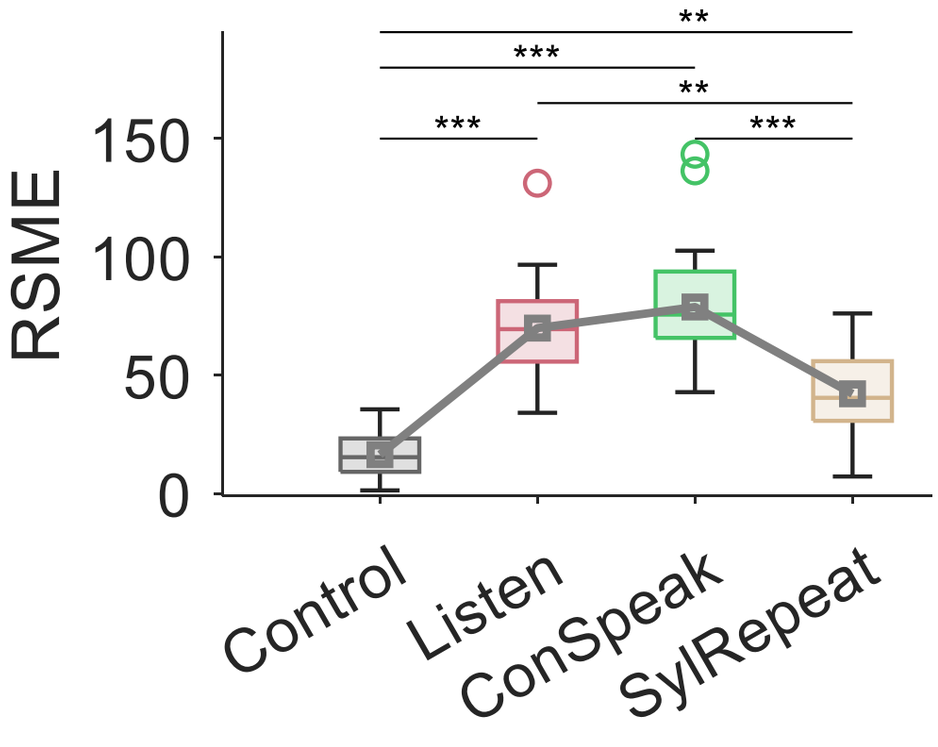
# BCI Performance




# Subjective Evaluation

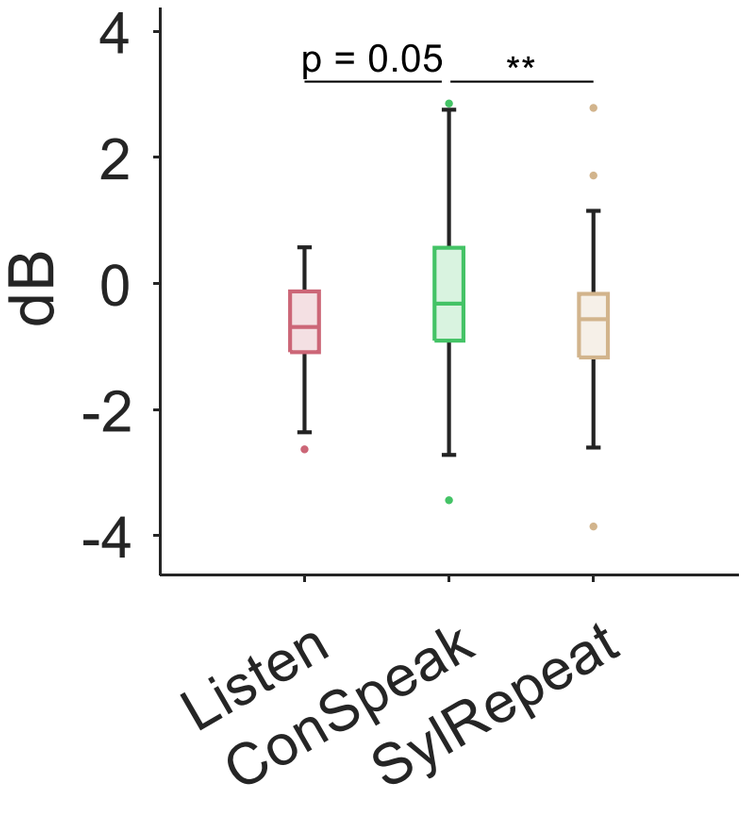


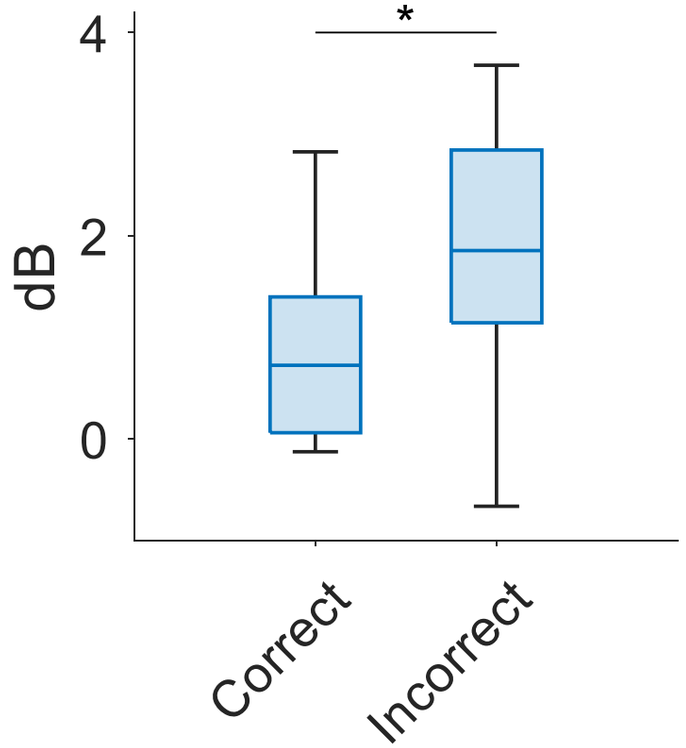



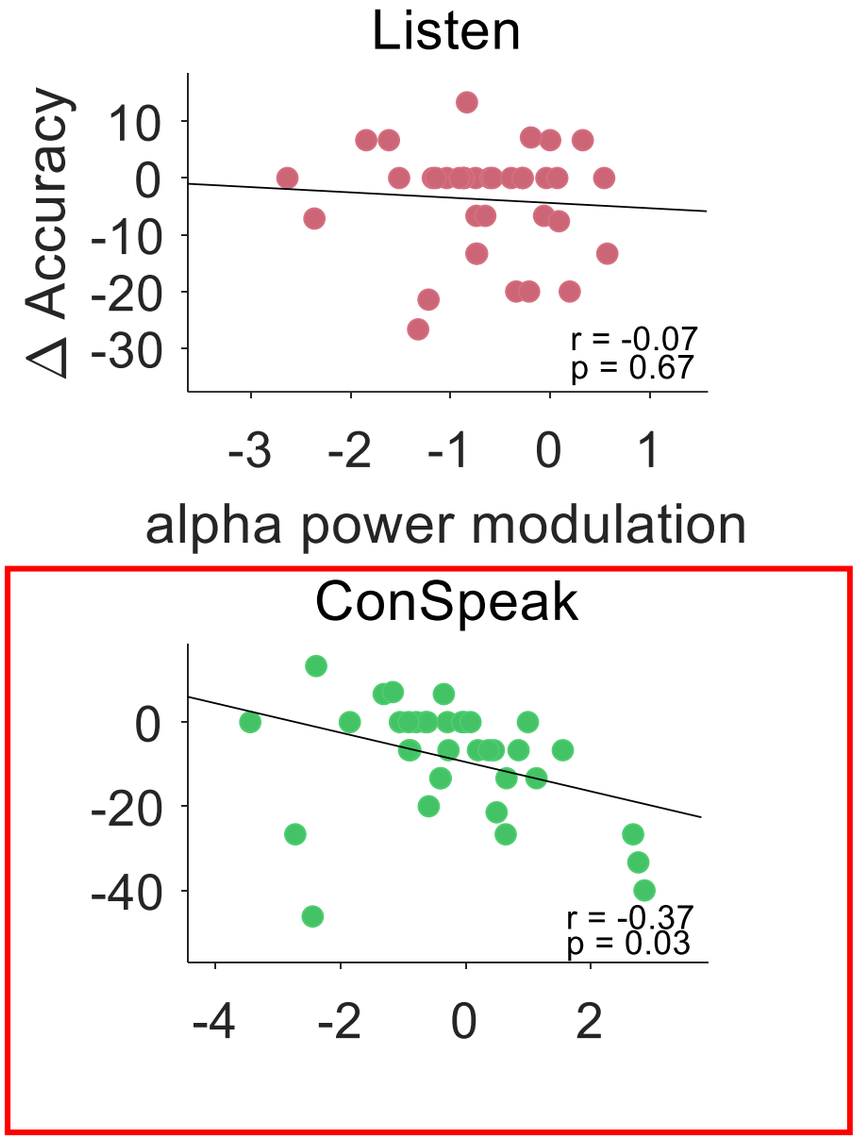

$p < 0.05$

(Kim & Kim, 2024)

# Alpha Power Modulation - Visual attention



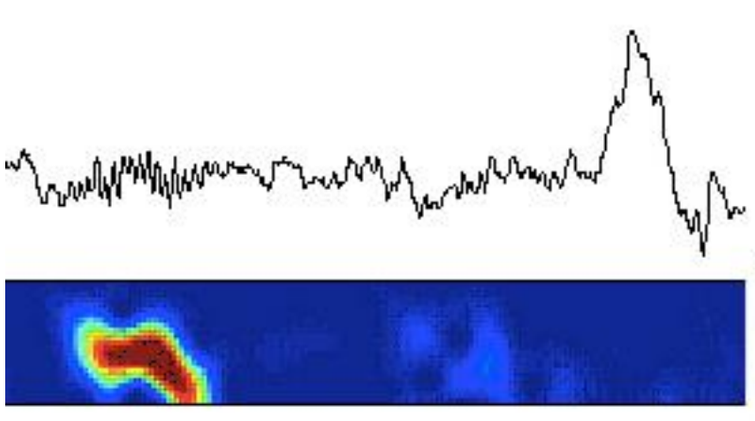

* $p < 0.05$

(Kim & Kim, 2024)

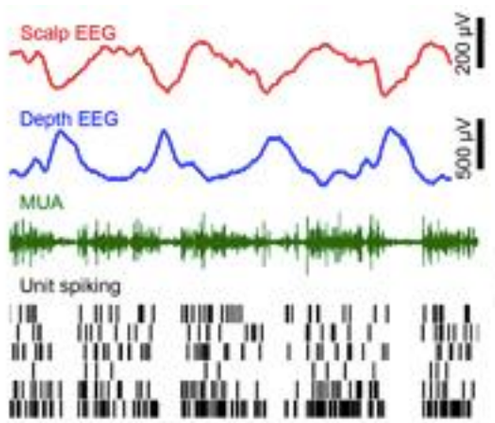
# Outline

- EEG Preprocessing
- Event-Related Potential (ERP)
- Brain Oscillation

# Brain Oscillations







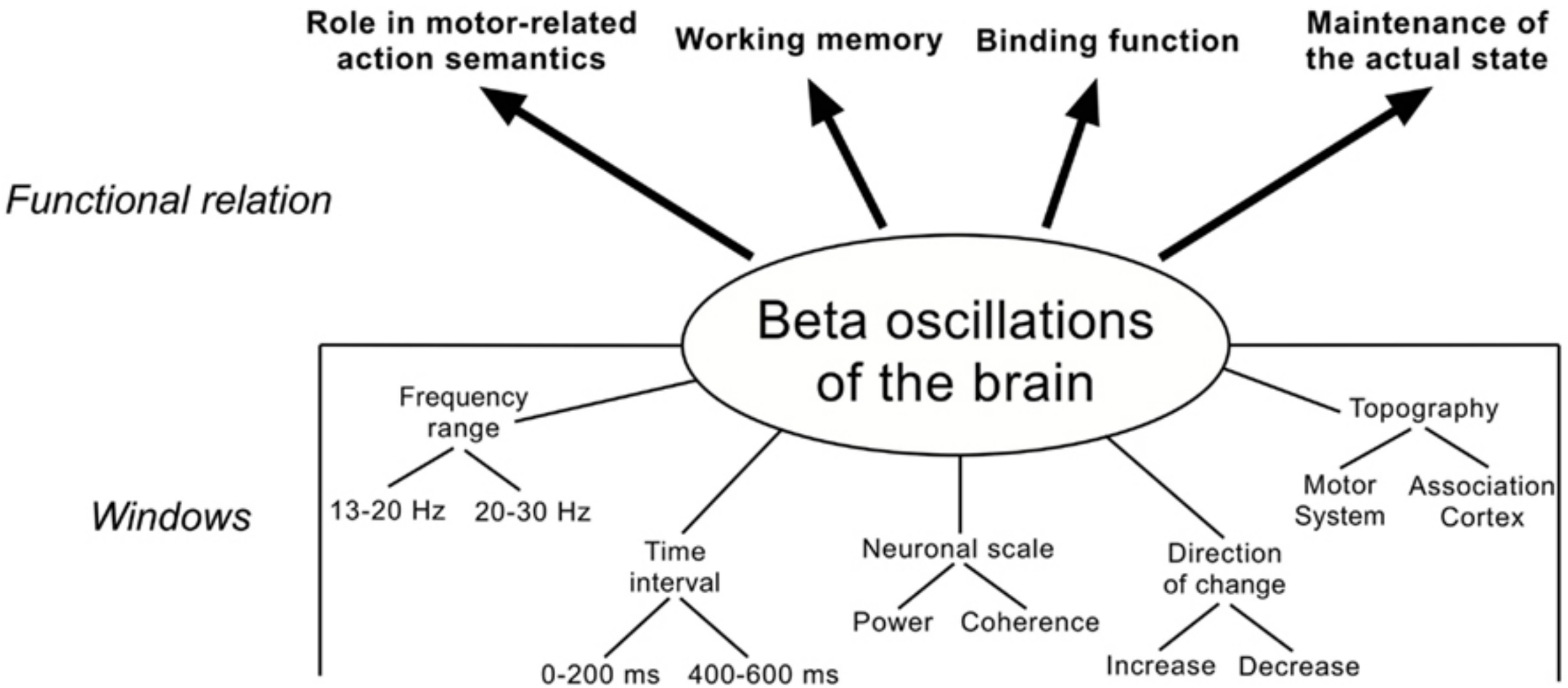




| Class[a] | Frequency band/Hz | Associated behaviors |
|---|---|---|
| δ | 1–4 | Slow wave sleep |
| θ | 4–8 | Spatial navigation learning (rats), REM sleep |
| α (also μ, κ) | 8–12 | Quiet wakefulness, somatosensory processing |
| β | 13–30 | Active wakefulness, motor control |
| γ | 30–70 | Spatial navigation learning, attention, perceptual binding |

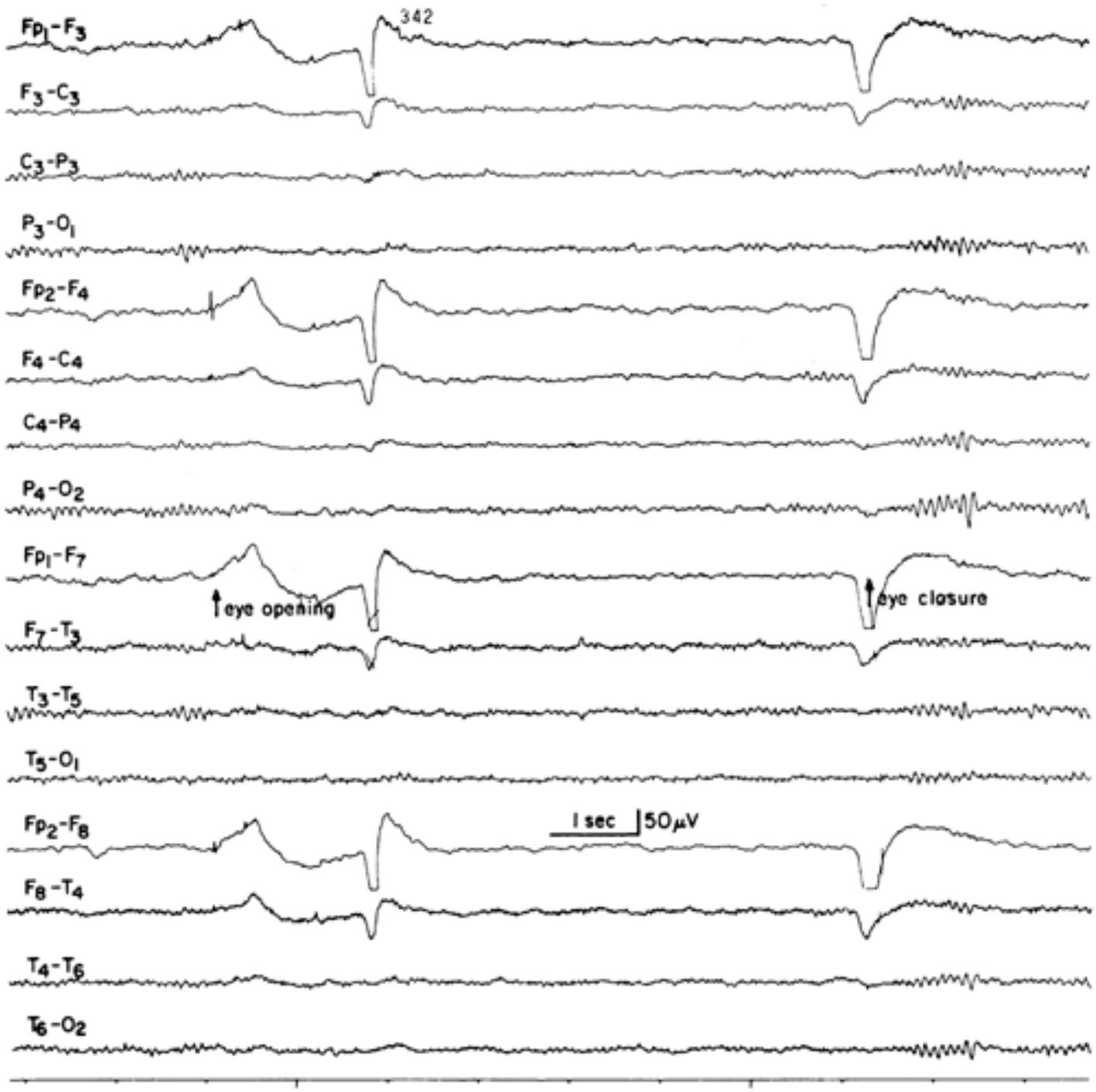
(Source: expertmind.com)

# Example




(Weiss and Muller, 2012)

# Alpha rhythm



# Fourier Transform

- Synthesis

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{jw})e^{jwn}dw$$

- Analysis

$$X(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n]e^{-jwn}$$

This is in fact called a discrete-time Fourier transform (DTFT)

# Discrete Fourier Transform (DFT)

- Given a sequence of a length N, x[n], the DFT of x[n] is,

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)kn} \qquad k = 0, \ldots, N\text{-}1$$

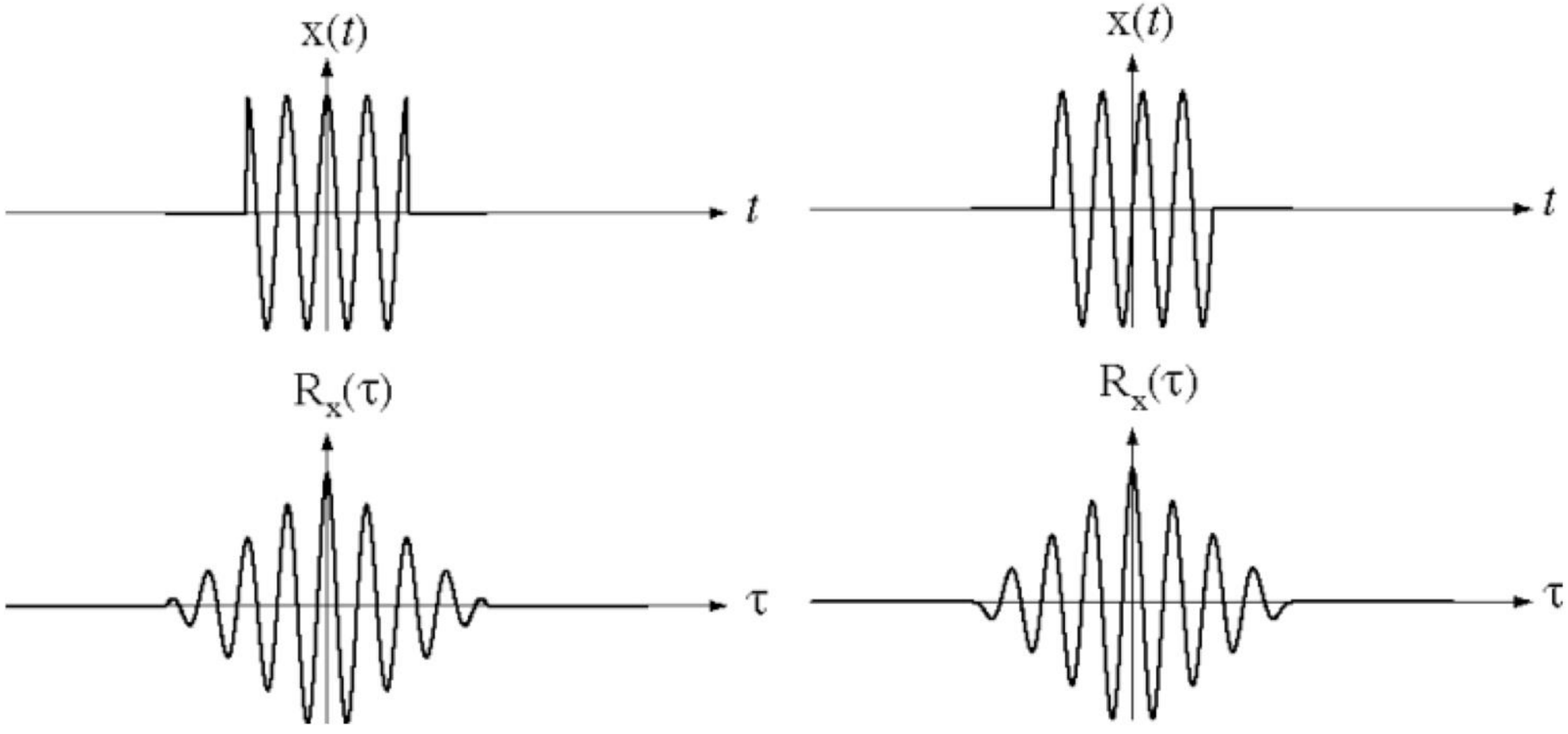
$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(2\pi/N)kn} \qquad n = 0, \ldots, \text{N-1}$$

# Fast Fourier Transform (FFT)

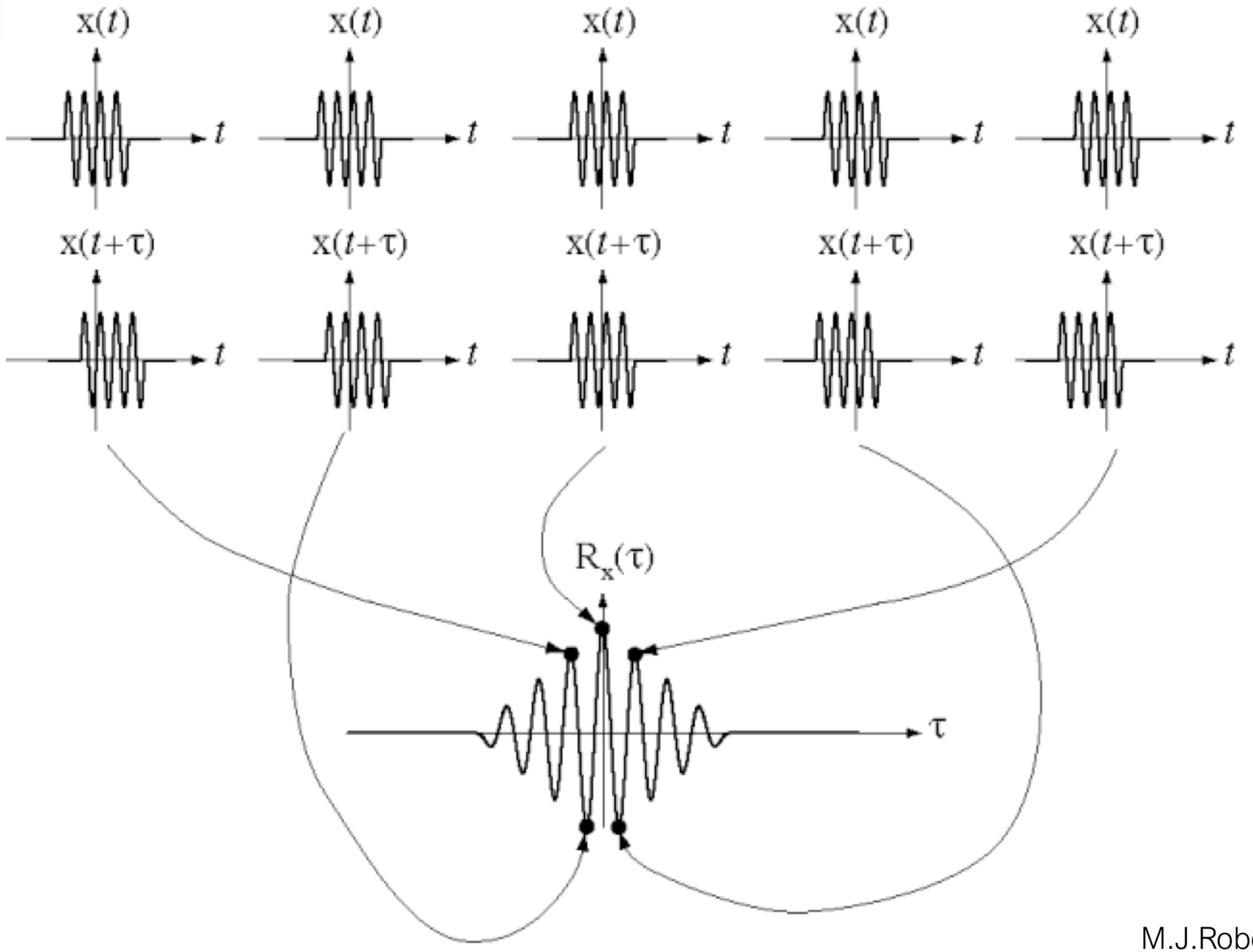
- Fast algorithms to compute the DFT of a signal
- Resize a given data to have the length of the power of 2 (then, we can split the data into half and half) for the best condition
- Either truncate the data or zero-pad to the end of the data

# Autocorrelation of a signal

$$R_{xx}[m] = \sum_{n=-\infty}^{\infty} x[n]x[n+m]$$




M.J.Roberts




M.J.Roberts

# Power Spectral Density

The Fourier transform of the autocorrelation of the signal x(t)

$$S_x(f) = \int_{-\infty}^{\infty} R_x(\tau)\, e^{-2\pi i f \tau}\, d\tau$$

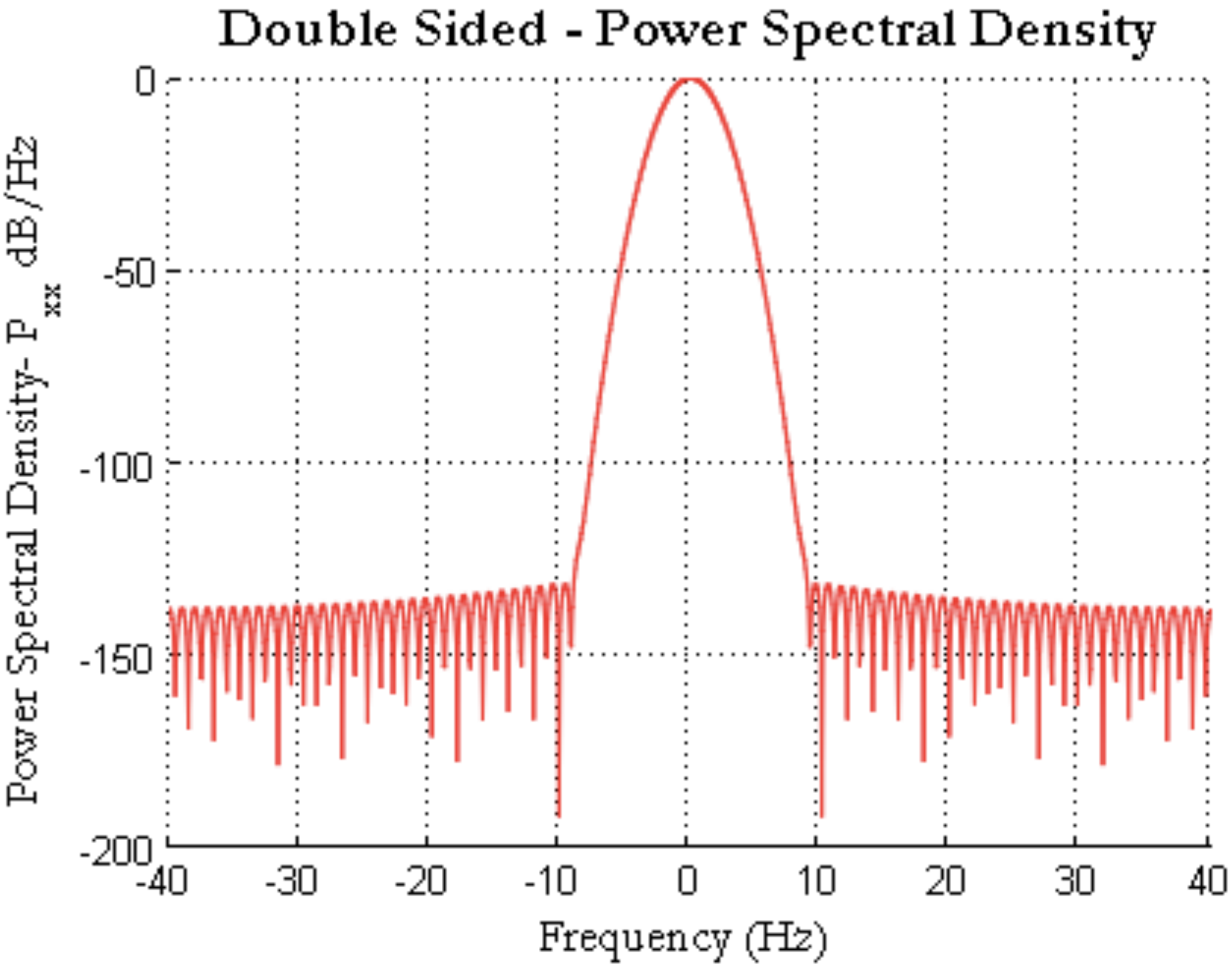
**How comes it describe the power?**

- We can derive the following equation:

$$S_x(f) = \lim_{T\to\infty} \frac{1}{T} E[\, |X_T(f)|^2 \,]$$

unit: $(SU^2)/Hz$

T: signal range


Double Sided - Power Spectral Density
Power Spectral Density- P xx dB/Hz
0
-50
-100
-150
-200
-40
-30
-20
-10
0
10
20
30
40
Frequency (Hz)

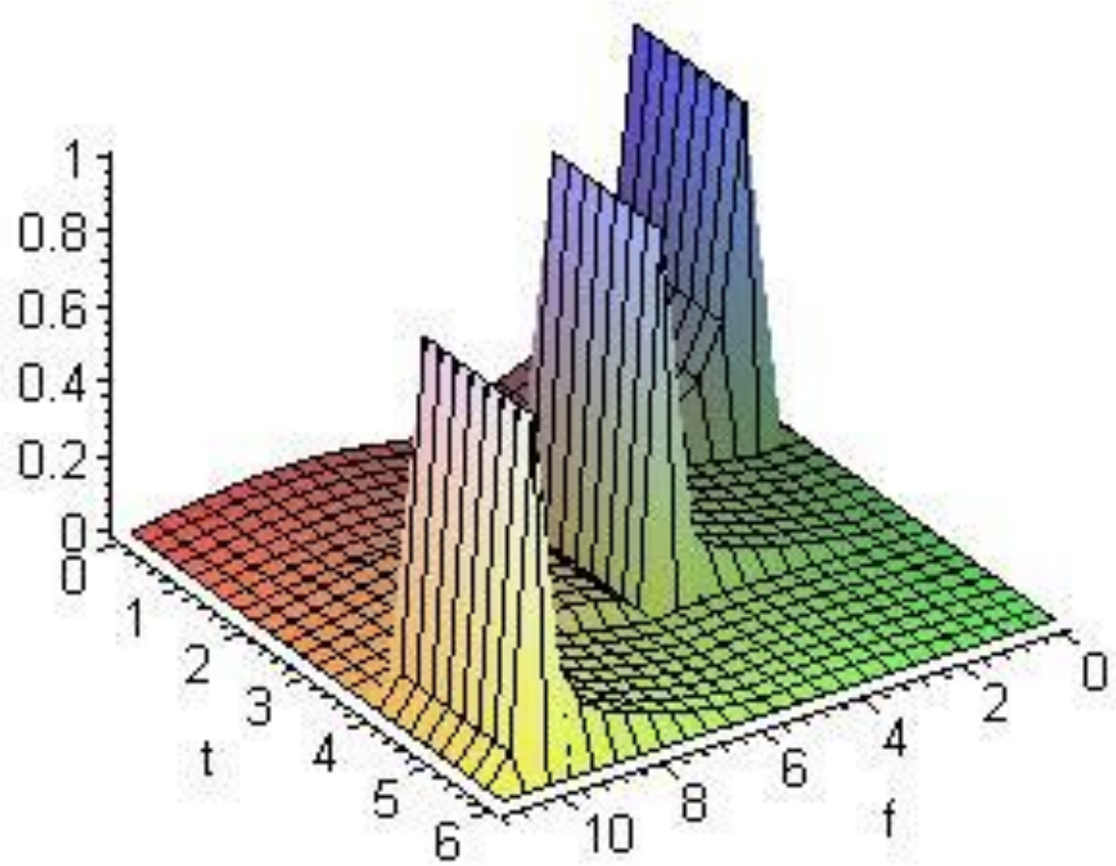
# Frequency analysis of non-stationary signals

- (Wide-Sense) Stationary signal
  - Mean does not vary over time
  - Auto-Correlation depends only on a time lag
    - Power spectral density remains the same over the entire signal
- What if the signal is non-stationary?
  - Need a time-varying frequency analysis

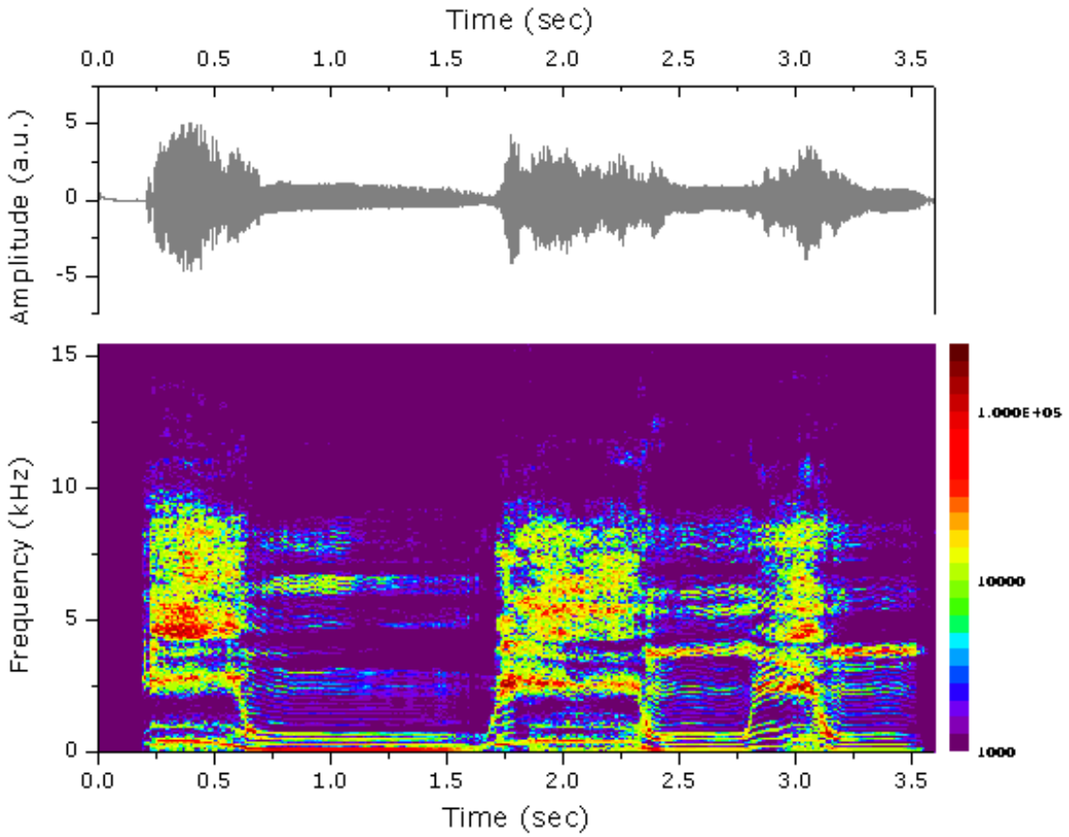
# Short-Time Fourier Transform (STFT)

- If a signal is stationary in a small time interval, the application of the FT in that interval works properly

- Approach
  - Select a segment with a short-time interval
  - Select a window function
  - Perform the DFT of the windowed signal in the segment
  - Slide the window forward in time with or without the overlap and repeat the computation of the STFT




Source: E. Aboufadel




Source:OriginLab

# Short-Time Fourier Transform (STFT)

- Procedure

1. Extract the $m$-th segment of the input signal, $x[n]$.

   $x_m[n] = x[n+mR]$, for n = -H, ..., 0, ..., H.  $R$ is the hop size.

   The segment length is $M = 2H + 1$ ($M$ is assumed to be odd in general)

2. Multiply $x_m[n]$ by the window function, $w[n]$.

   $y[n] = x_m[n]\, w[n]$

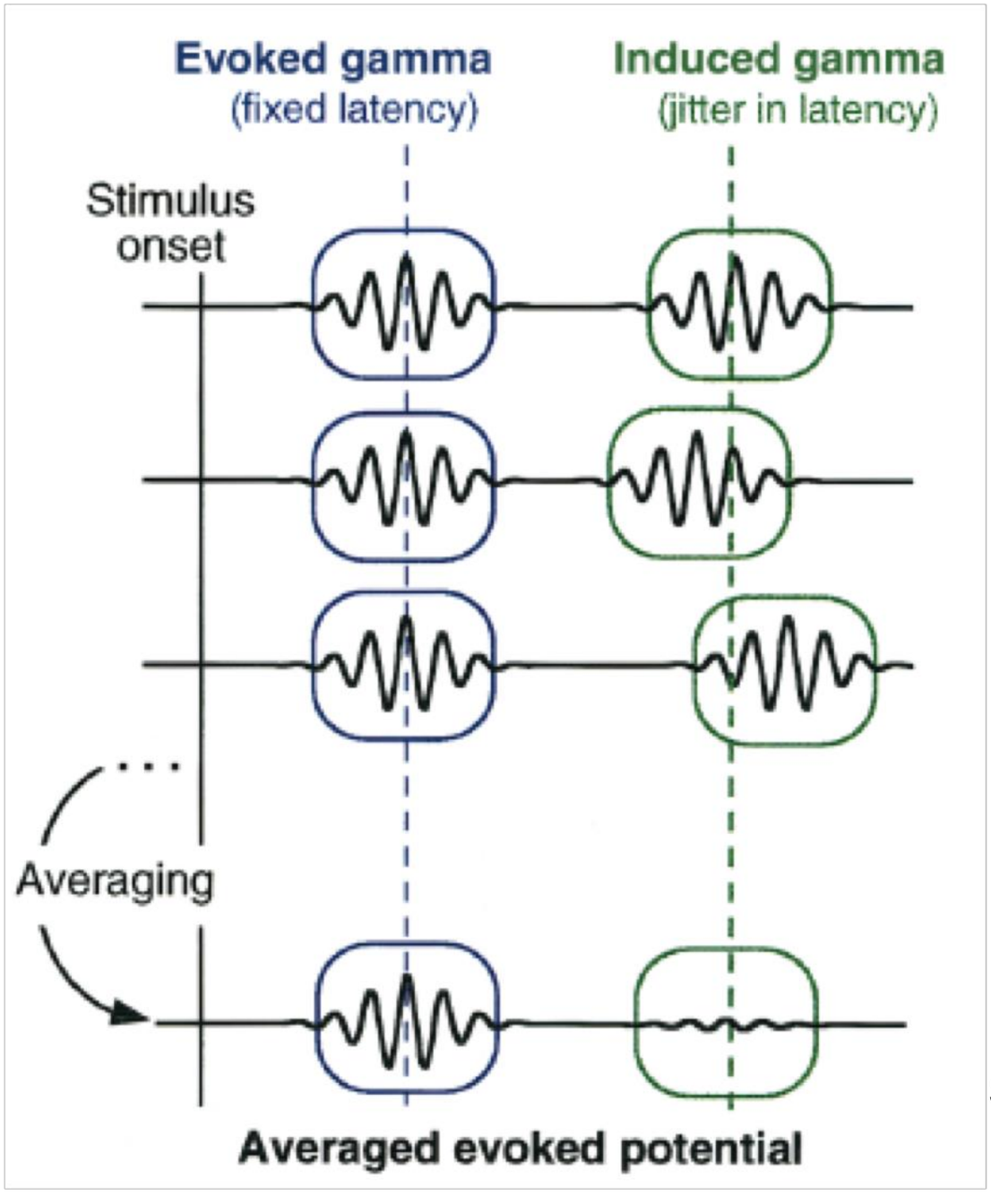
3. Zero-pad the windowed data to have the size of $N$ (a power of 2)

   $z[n] = y[n]$ if $|n| \leq H$, or 0 if $H < n \leq N/2-1$, or 0 if $N/2 < n < -H$

4. Take the FFT of the zero-padded data

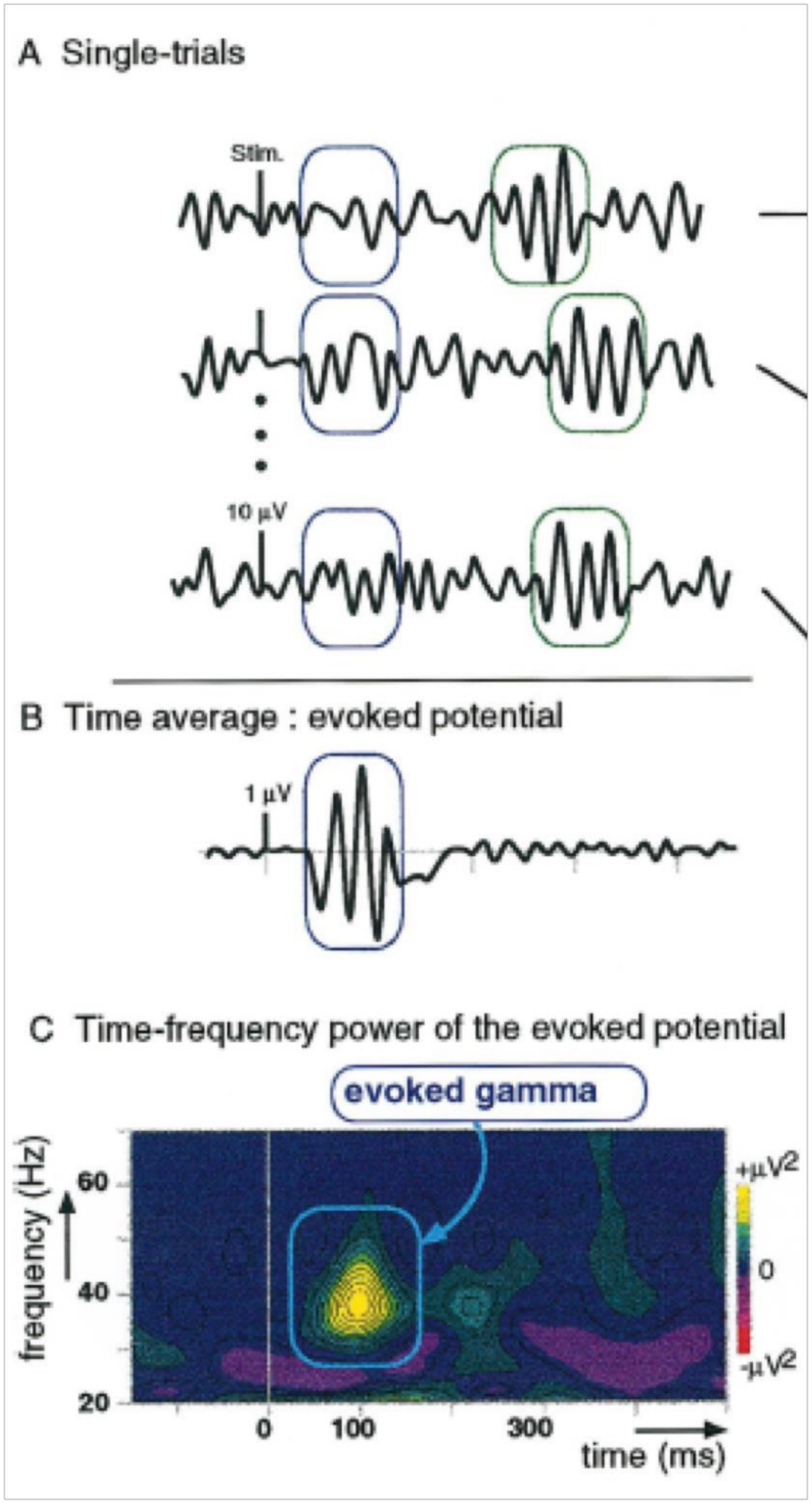
5. Increase $m$ to $m+1$ and repeat the steps 1-4

# Evoked vs. Induced Brain Oscillations

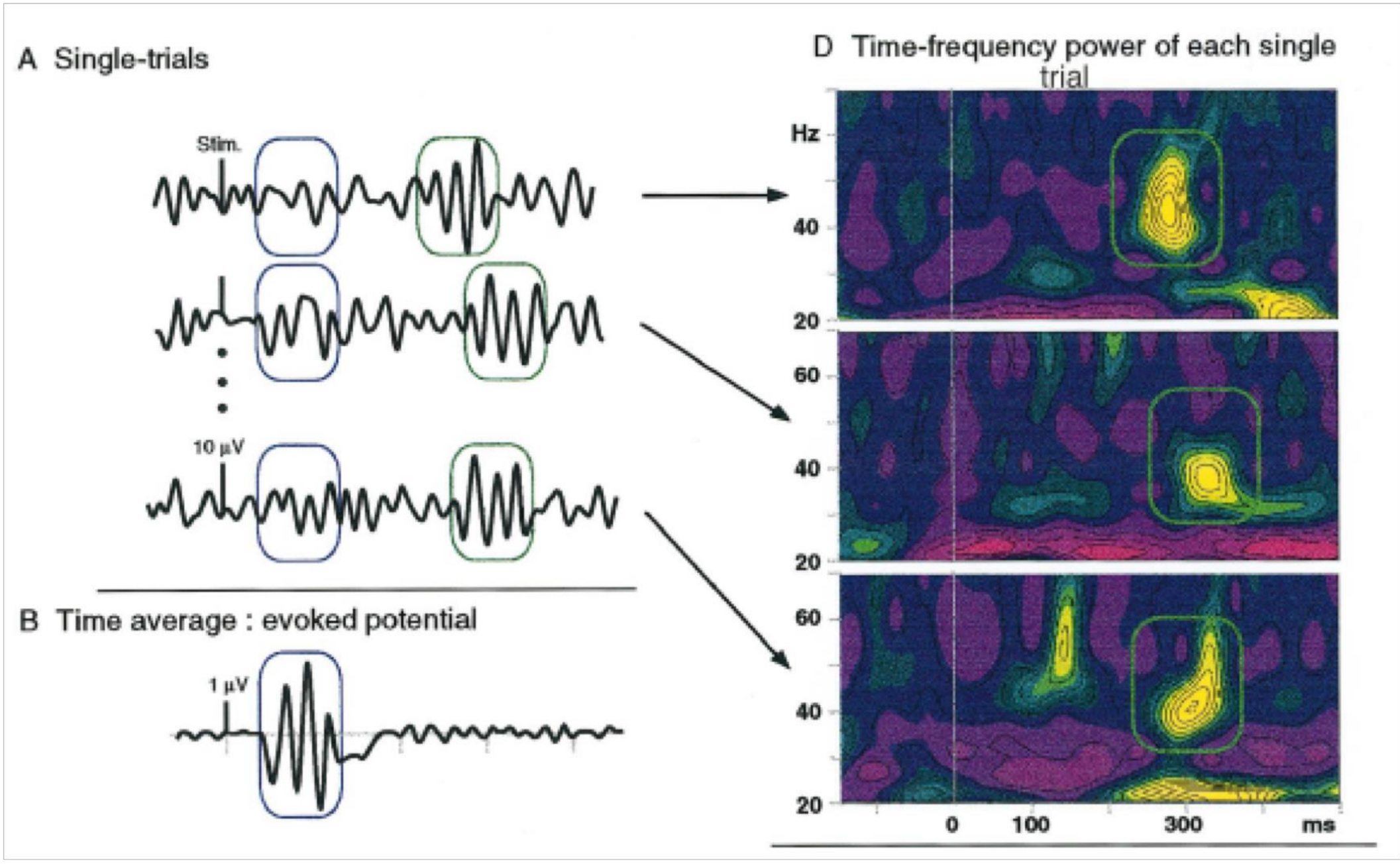



Source: Tallon-Baudry and Bertrand

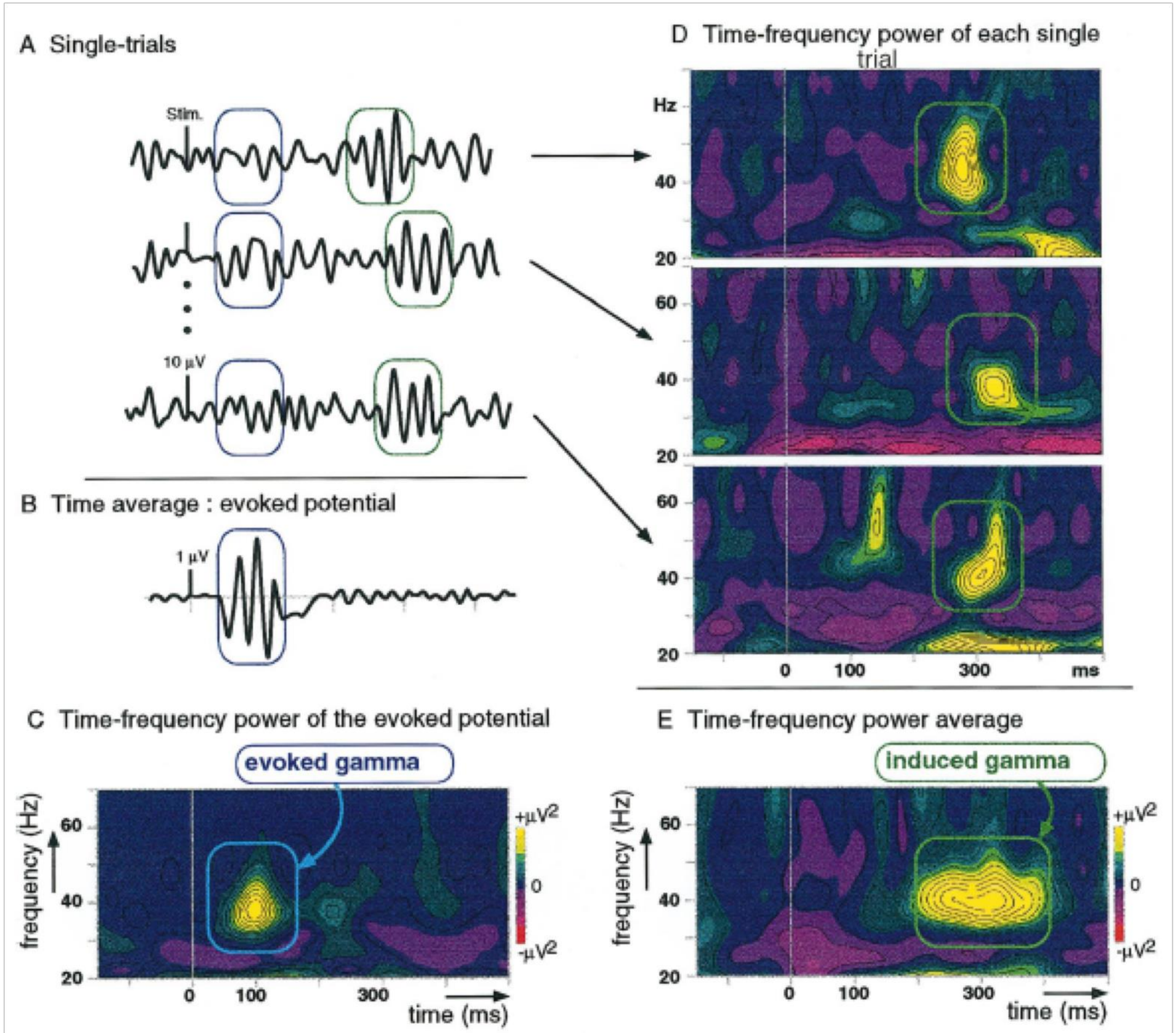
# Evoked Gamma Activity



# Induced Gamma Activity

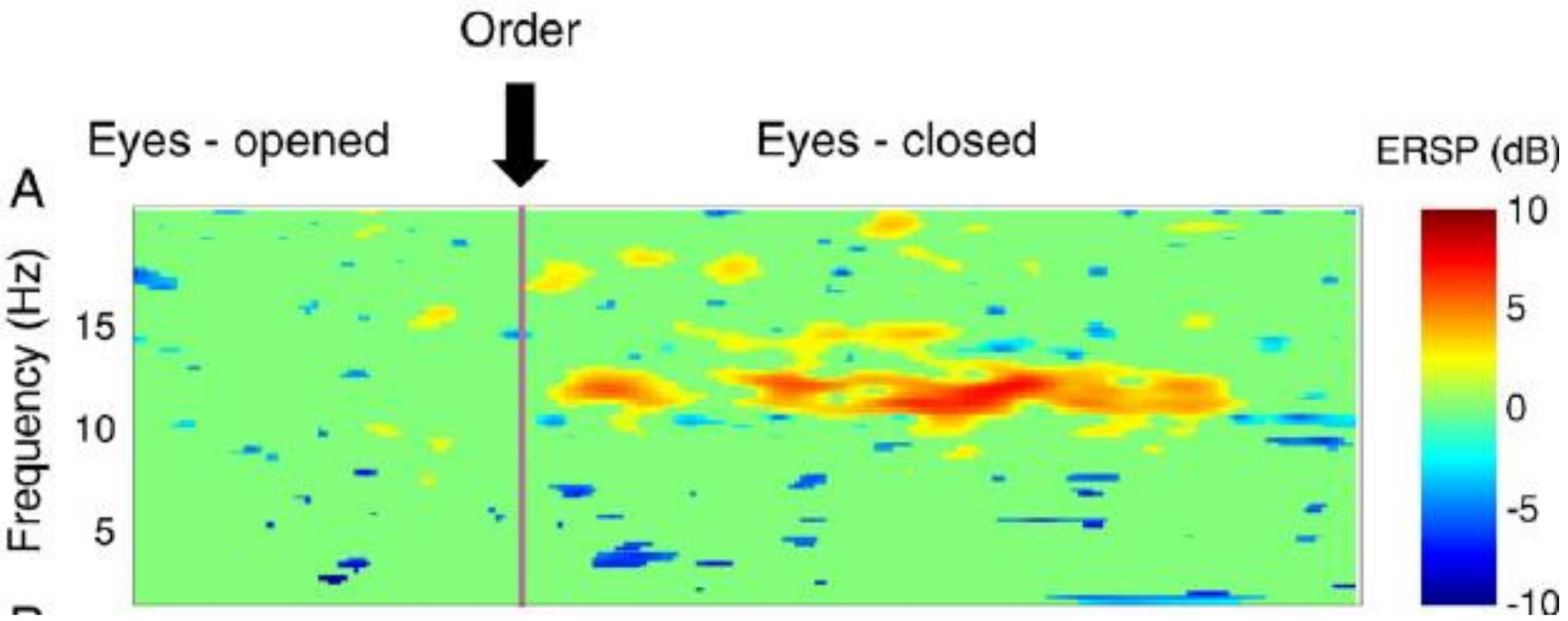


# Induced Gamma Activity



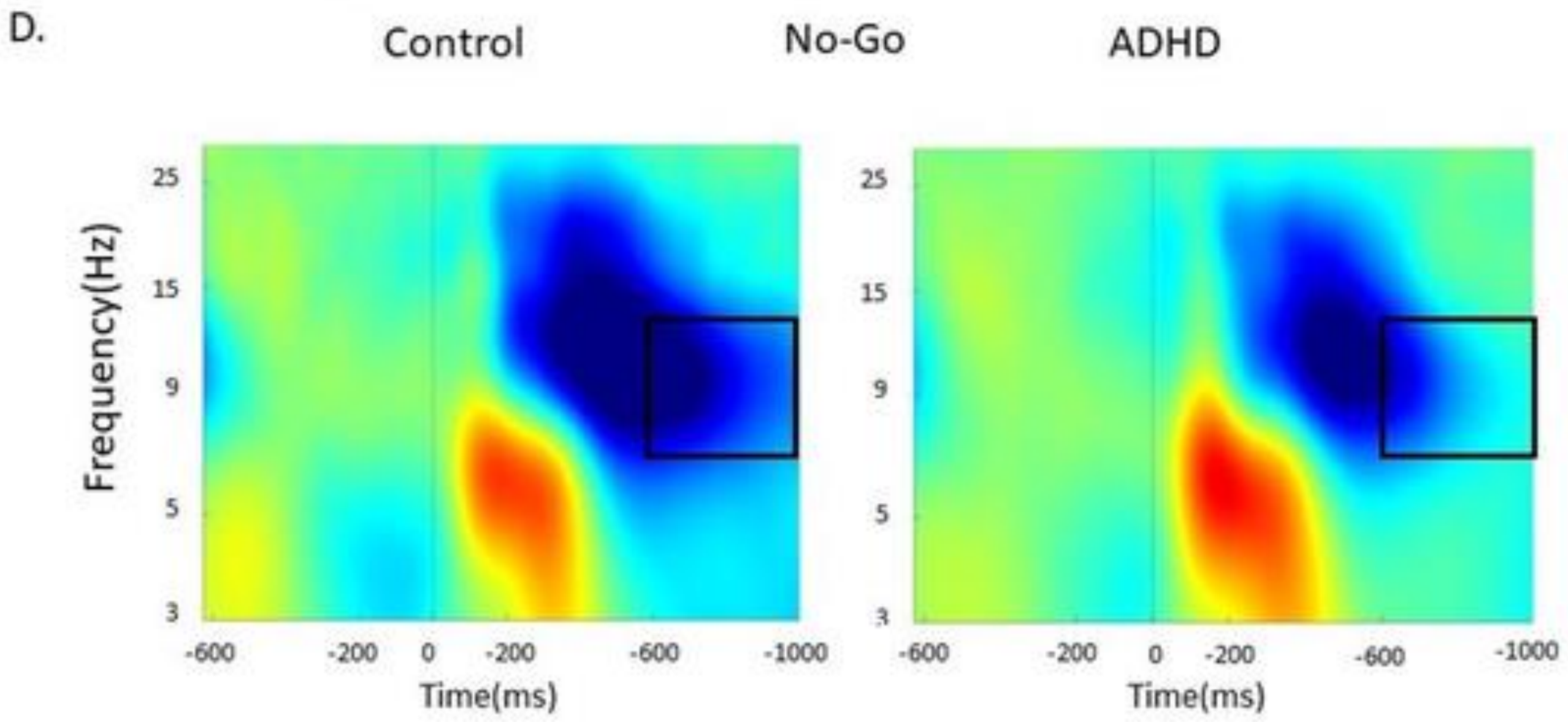
# Event-Related Spectral Perturbation (ERSP)

- Measurement of average changes in EEG spectrum over time induced by events




(Cheron et al., 2006)

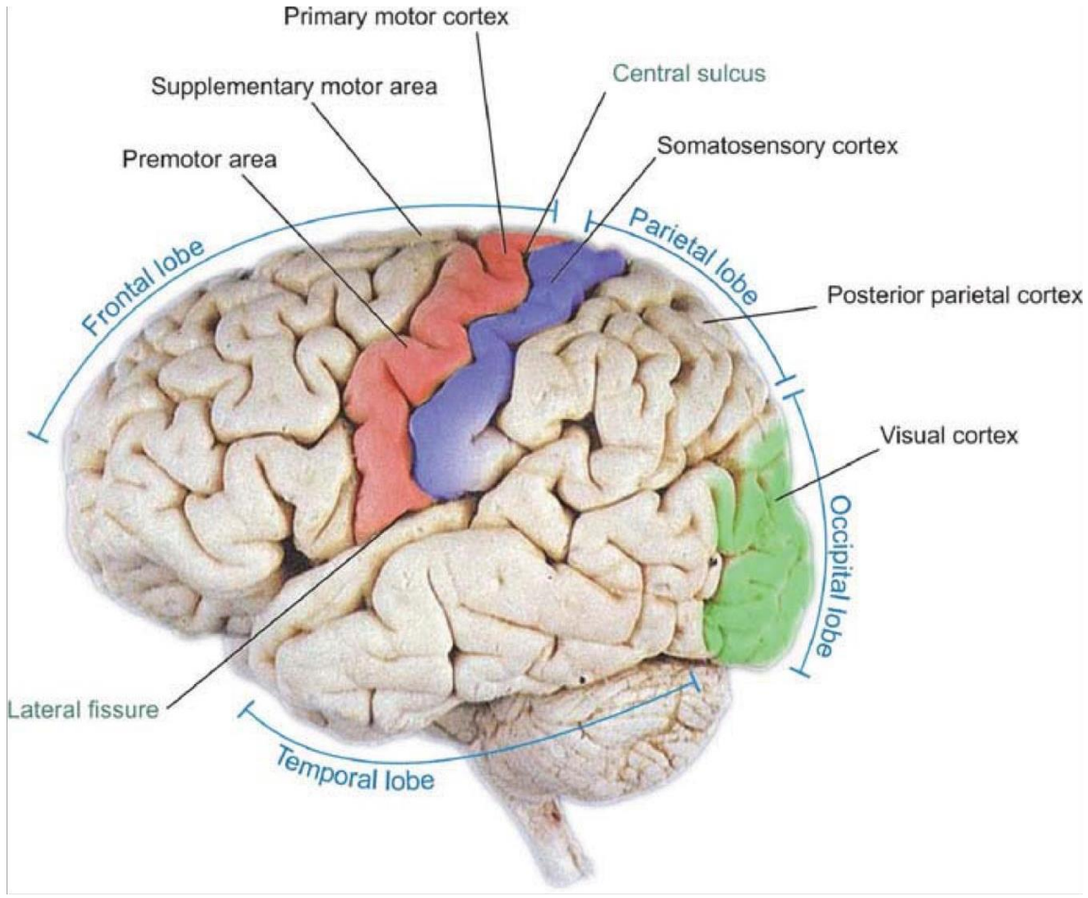
# Event-Related Spectral Perturbation (ERSP)




(Bozhilova et al., 2020)

# Case Study - Motor Imagery BCI

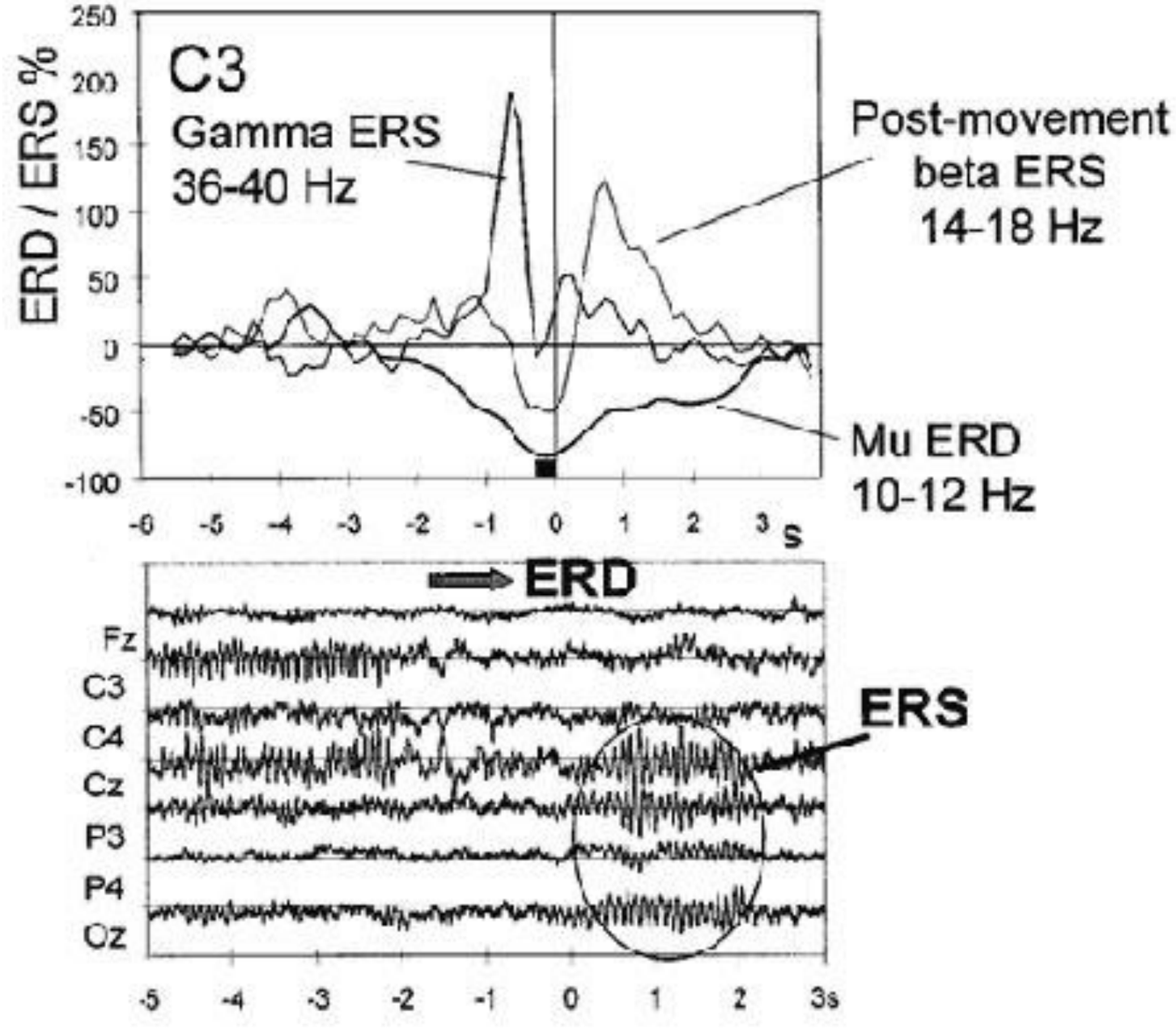
- Sensorimotor Rhythm (SMR)
  - Generated from senosorimotor area
    - Somatosensory cortex
    - Motor cortex
- Mu rhythm: 8-12 Hz
- Beta rhythm: 12-30 Hz



# Motor Imagery BCI

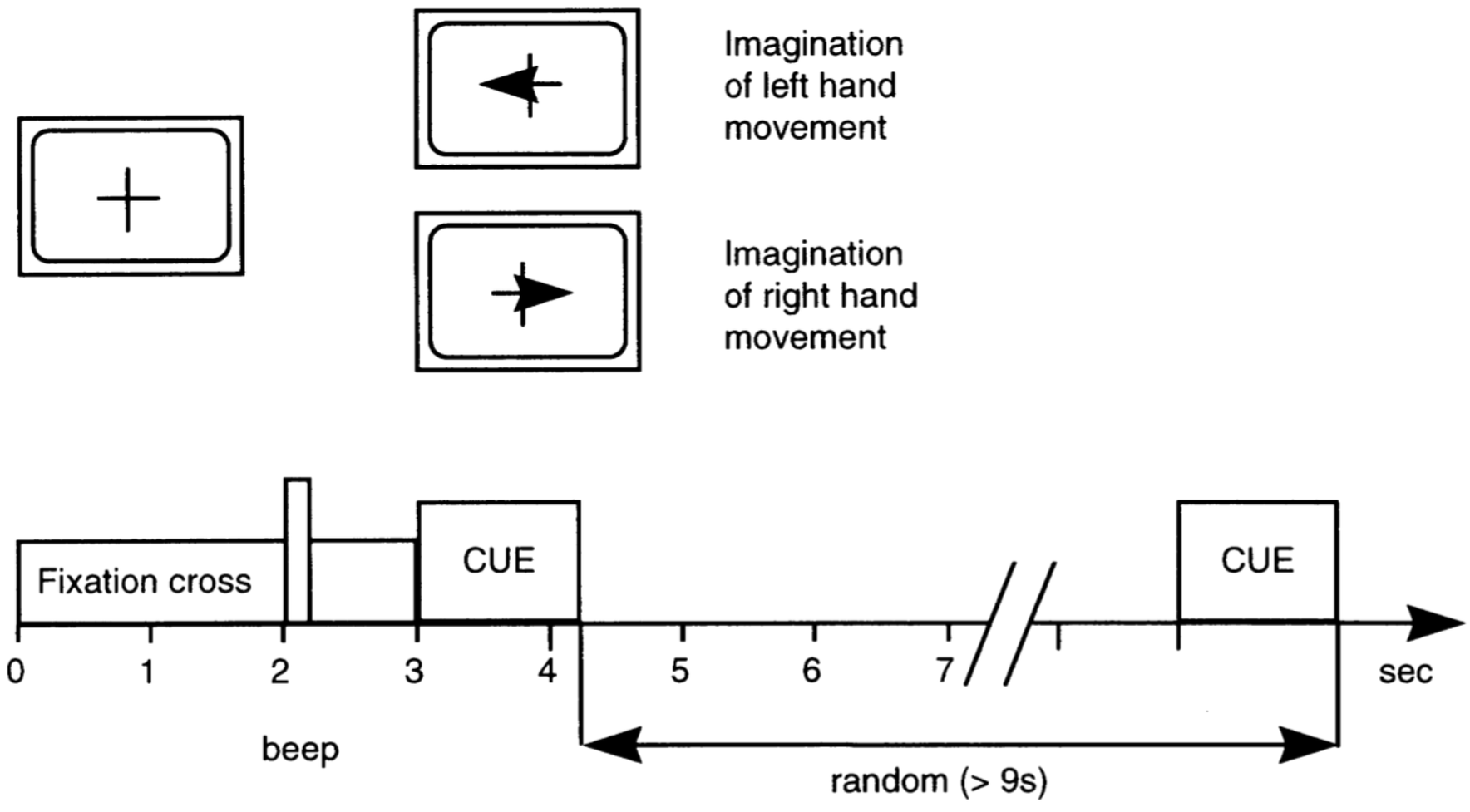
- Changes in SMR power related to movement
  - Motor planning and execution
    - SMR power decreases = event-related desynchronization (ERD)
  - Motor inhibition
    - SMR power increases = event-related synchronization (ERS)
- More pronounced at contralateral area
- Can be generated by imagined movement, too

# Motor Imagery BCI



# Motor Imagery BCI

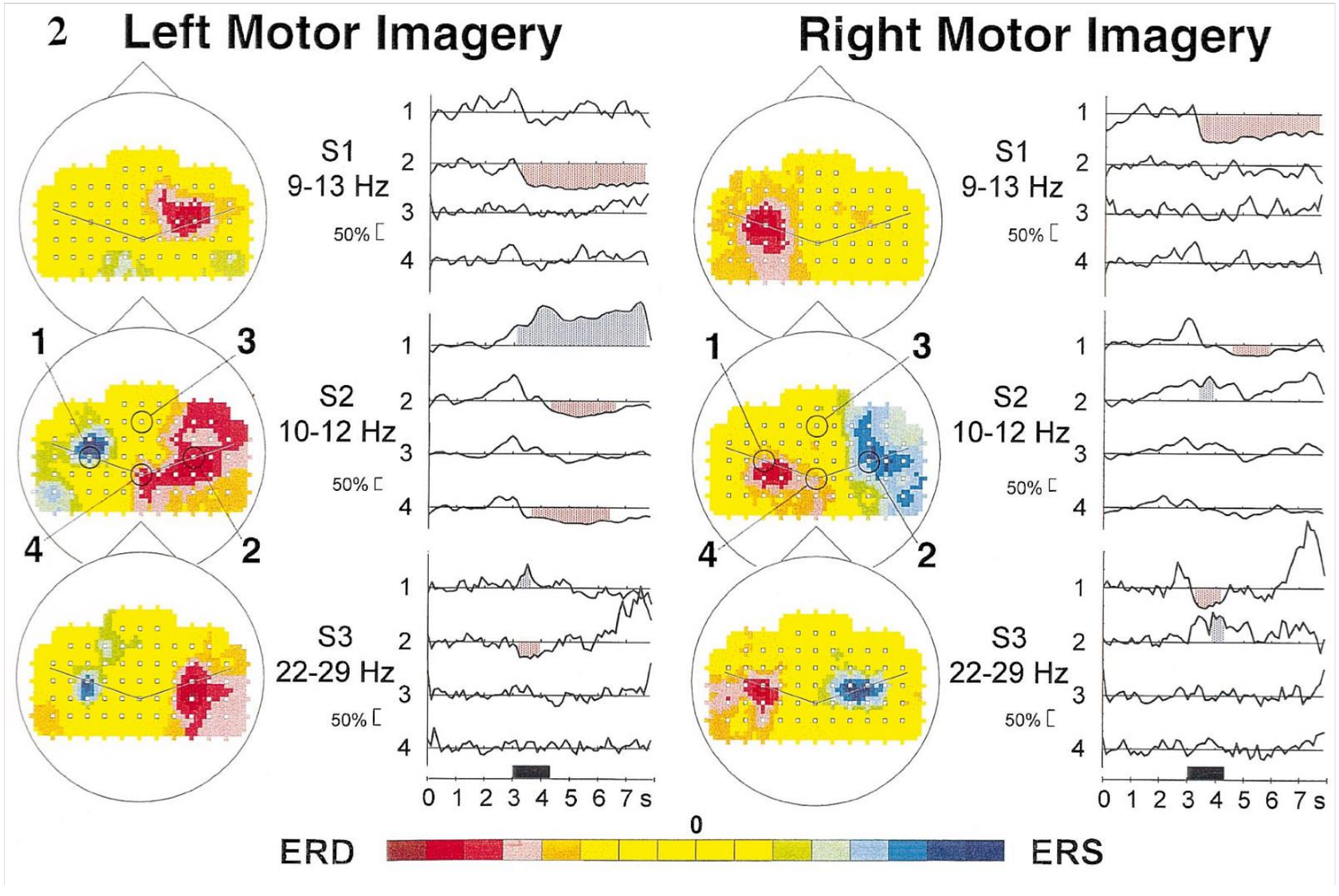
## SMR changes by imagined movement



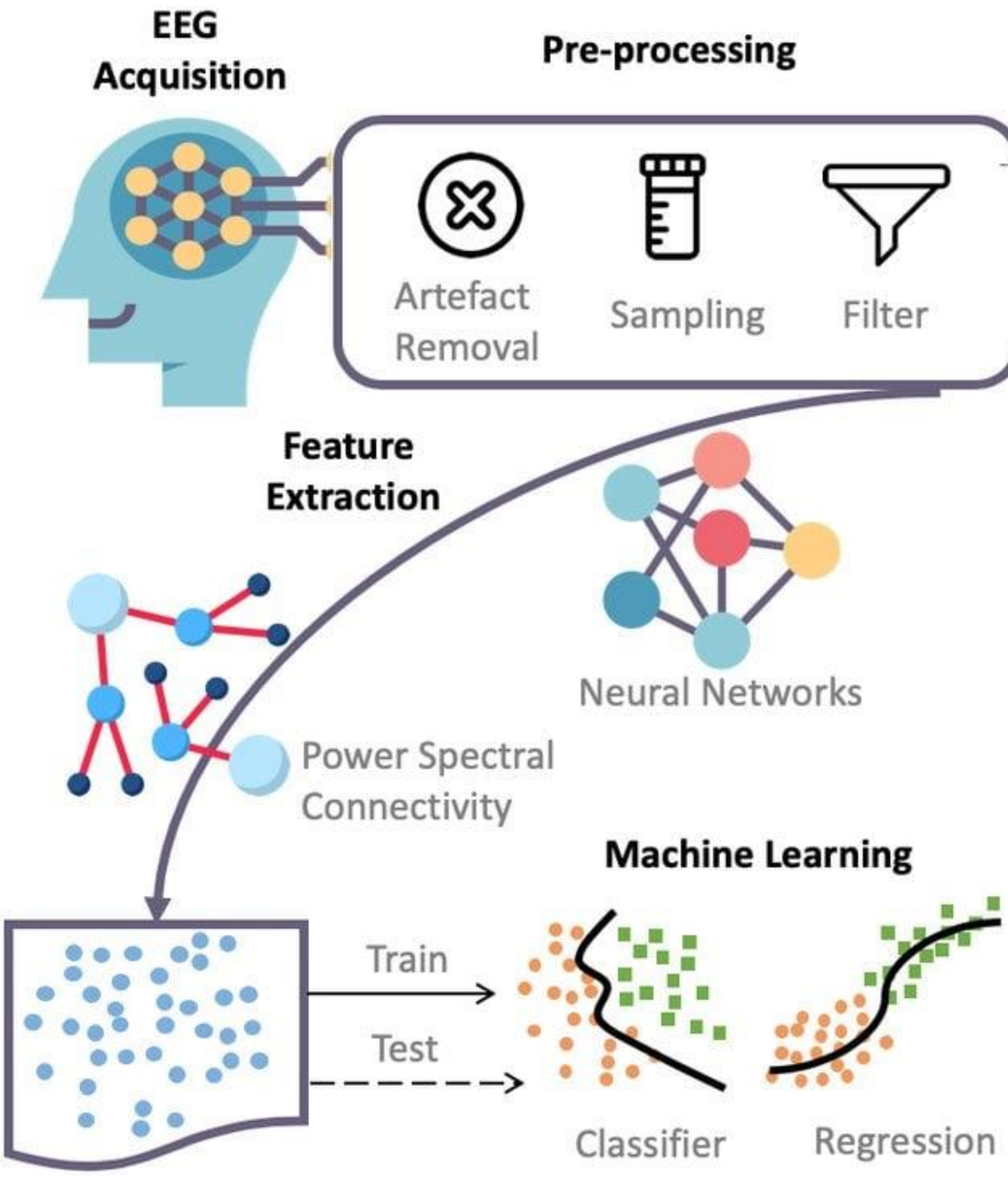

Pfurtscheller and Neuper, 1997

# Motor Imagery BCI

SMR changes by imagined movement




Pfurtscheller and Neuper, 1997


EEG
Acquisition
Pre-processing
Artefact
Removal
Sampling
Filter
Feature
Extraction
Neural Networks
Power Spectral
Connectivity
Machine Learning
Train
Test
Classifier
Regression

(Gu et al., 2020)